

Rewriting the Universe

High-Performance Hypergraph Evolution for the Wolfram Physics Project

Richard Assar

Wolfram Institute

rassar@wolfram.institute.org

August 31, 2026

Abstract

The Wolfram Physics Project models fundamental physics by hypergraph rewriting, and its central computation, exhaustive multiway evolution, grows exponentially with depth. We present an engine for it. States are identified exactly: a McKay-style individualization–refinement canonical hash is computed for every state, by one implementation shared between the CPU and, executed warp-collectively, the GPU’s resident evolution kernel; a proved ceiling bounds what any cheaper filter placed in front of it could save. The engine is lock-free and non-phasic throughout, the causal and branchial relations and their transitive reduction maintained online, and its output is a function of the inputs alone, independent of thread count and scheduling. Quotient exploration expands one representative per isomorphism class and reconstructs the raw events, causal edges and branchial pairs exactly; caps, transition sampling and session steering bound an exploration without a barrier. The shared concurrent protocols are model-checked under RC11 and scoped-RC11 and the forwarding rendezvous under TLA+, at stated bounds. Against the Wolfram/Multicomputation packet’s `MultiwaySystem`, computing identical state and causal-edge counts at every depth, the engine is faster by $2,770\times$ at depth 4 and $30,627\times$ at depth 7. Every measurement in this report is generated by a named tool whose provenance line records the commit and machine. The system is available as a C++/CUDA library and as a Wolfram Language packet.

Keywords: hypergraph rewriting, graph canonicalization, multiway systems, GPU computing, Wolfram Physics Project

1 Introduction

The Wolfram Physics Project [1] proposes that the fundamental structure of the universe emerges from the repeated application of simple rewriting rules to hypergraphs. This approach generates three derived structures: causal graphs, recording dependencies between rewriting events; branchial graphs, connecting states reachable at the same evolution step; and multiway evolution graphs, representing the set of all reachable

histories.

Exhaustive multiway evolution is computationally demanding. The number of reachable states grows exponentially with the number of evolution steps, and each step requires pattern matching against the current hypergraphs, canonicalization to identify isomorphic duplicates, and event recording to construct the causal and branchial structures. The existing Wolfram Language implementation is mathematically complete, but slow enough that its running time rather than the definition sets what can be explored. Section 8 measures it at 3.7s to reach the six canonical states of depth five on the corpus’s `wolfram-2to4` rule, against 0.3ms for the engine on the same evolution, and the gap widens with depth.

This paper presents a high-performance implementation addressing these challenges through:

1. **Exact canonicalization:** We use McKay-style individualization–refinement (IR) [2] as the canonicalizer, and under the exact mode it runs on every state: its hash is the deduplication key, so two states are identified if and only if they are isomorphic. A Weisfeiler–Leman signature [3] serves the approximate modes, which exist for callers that ask for a coarser identity. Proposition 3.3 shows why it is not placed in front of IR: the saving available to such a filter is bounded above by the ratio of canonical classes to raw states, which falls toward zero as an evolution deepens.
2. **Lock-free concurrency:** The engine holds no lock. Its concurrent hash maps, segmented arrays, lock-free lists and work-stealing dequeues are non-blocking, and a worker with no work waits on an address rather than spinning.
3. **GPU acceleration:** A CUDA backend runs matching, rewriting and canonicalization in one kernel that stays resident across evolution steps, so no step returns to the host. It is faster than the host where the rule set yields enough independent work to fill the device, which Section 8 quantifies.
4. **Configurable equivalencing:** State and event equivalencing are independent axes. State canonicalization is selected as NONE, AUTOMATIC (fast

content hash), or FULL (exact IR); event canonicalization is composed from a small set of signature keys. There is no user-facing choice between competing canonicalization algorithms—the exact path is always IR, computed on every state.

5. **Exploration strategies:** Quotient exploration (Section 5.10) expands one representative per isomorphism class while retaining the full event set; transition-level sampling with per-rule weights (Section 5.13) admits runs at depths where exhaustive expansion is not available; and the causal graph’s transitive reduction (Section 5.11) is maintained online rather than as a pass over the finished graph.
6. **A determinism contract:** The state set, event set and causal and branchial relations are a function of the inputs alone, independent of thread count and scheduling order (Section 5.8).

The remainder of this paper is organized as follows. Section 2 reviews hypergraph rewriting and the Wolfram Physics framework. Section 3 presents our canonicalization algorithms. Section 4 describes the pattern matching system. Section 5 details the system architecture. Section 6 covers GPU acceleration. Section 8 presents experimental results. Section 9 concludes with future directions.

2 Background and Related Work

This section fixes the objects the rest of the paper measures and states which of their properties are the expensive ones. Hypergraph rewriting supplies the rule and the match; multiway evolution supplies the branching structure that makes a state space rather than a trajectory, together with the causal and branchial relations derived from it. The cost of the whole construction concentrates in one place: to know whether two states are the same state, the system must decide hypergraph isomorphism, and it must decide it once per produced state rather than once per query. We close with the systems and libraries this work is measured against.

2.1 Hypergraph Rewriting Systems

A *hypergraph* $\mathcal{H} = (V, E)$ consists of a vertex set V and a set of hyperedges E , where each hyperedge $e \in E$ is an ordered tuple of vertices $(v_1, v_2, \dots, v_k)$ with $v_i \in V$. Unlike standard graphs where edges connect exactly two vertices, hyperedges may connect any number of vertices with arbitrary arity.

Definition 2.1 (Rewriting Rule). *A rewriting rule $\mathcal{R} = (L, R)$ consists of a left-hand side pattern L and right-hand side replacement R , both hypergraphs. A match of L in $\mathcal{H}$ is an injective mapping $\phi : V_L \rightarrow V_{\mathcal{H}}$ such that the image of each hyperedge in L appears in $\mathcal{H}$.*

This is the single-pushout form of rewriting: the rule is applied by deleting the matched image and gluing in the replacement, without the additional gluing conditions that the double-pushout construction imposes [4–6]. Applying rule $\mathcal{R}$ at match ϕ produces a new hypergraph $\mathcal{H}'$ by:

1. Removing all hyperedges in $\phi(L)$ from $\mathcal{H}$
2. Adding hyperedges from R , extending ϕ to map new vertices in R to fresh vertices

2.2 Multiway Evolution

Multiway evolution explores all possible rule applications simultaneously, generating a directed acyclic graph of states connected by events. Figure 1 shows two steps of a small rule in full—the raw states, the canonical states they collapse to, the events between them and the causal edges—and is the smallest complete instance of everything the rest of the paper computes.

Definition 2.2 (Evolution Graph). *Given initial state S_0 and rule set $\{\mathcal{R}_i\}$, the evolution graph $G = (\mathcal{S}, \mathcal{E})$ contains:*

- *States $\mathcal{S}$: all hypergraphs reachable from S_0*
- *Events $\mathcal{E}$: transitions (s, s', r, ϕ) where state s rewrites to s' via rule r at match ϕ*

The *causal graph* captures dependencies between events based on shared hyperedges. The *branchial graph* connects states reachable at the same evolution step, encoding branching structure analogous to quantum superposition [1].

2.3 The Graph Isomorphism Problem

State deduplication requires detecting when two hypergraphs are isomorphic—structurally identical up to vertex relabeling. Graph isomorphism is in NP and is neither known to be NP-complete nor known to be in P [7]; the best known bound is quasipolynomial [8].

For practical systems, we employ *graph hashing*: computing a canonical signature $h(\mathcal{H})$ such that isomorphic graphs have identical hashes. While perfect hashing (injective for non-isomorphic graphs) requires solving isomorphism exactly, approximate hashes suffice when combined with explicit isomorphism testing for collisions.

2.4 Related Work

Graph canonicalization: The nauty/Traces line of work [2] establishes individualization–refinement (IR) as the practical standard for exact graph canonicalization; it is exponential in the worst case but fast on typical instances, and it is the exact canonicalizer we adopt as our reference. Weisfeiler–Leman color refinement [3, 9] runs in polynomial time but provably fails to distinguish certain non-isomorphic graphs, so it cannot serve as an identity where exactness is required;

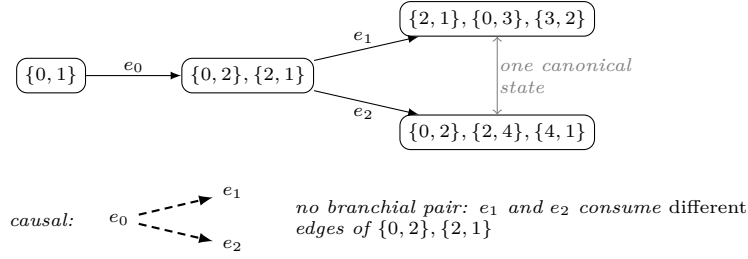


Figure 1: Two steps of $\{x, y\} \rightarrow \{x, z\}, \{z, y\}$ from a single edge, exactly as the engine reports it. Four raw states become three canonical ones: the two depth-2 results are isomorphic and are identified. Both causal edges descend from e_0 , because each depth-2 event consumes one of the two edges e_0 produced. The branchial relation is empty here, and for a reason worth stating: e_1 and e_2 share an input state, which resembles a sibling pair, but branchial edges connect co-consumers of a *common* edge and these consume different ones. Every element is output of `canonical_causal_oracle --dump 2`.

we use it only for the modes that ask for a deliberately coarser identity, never in front of IR.

Hypergraph rewriting implementations: SetReplace [10] is the implementation most closely associated with the Wolfram Physics Project, and is the closest system to this work. Throughout this paper, *reference implementation* denotes this project’s own brute-force evolver, named below, and never SetReplace. The two target different objects. SetReplace’s multiway construction is a token–event graph over a single growing token pool: it records which rewrites occurred and what each consumed and produced, and does not canonicalize or deduplicate states during evolution, so two structurally identical states reached along different histories remain distinct vertices. The object this engine enumerates is the states graph under an exact isomorphism identity, which requires canonicalizing every state as it is produced (Section 3); canonicalization therefore dominates the cost, rather than matching. The two objects are complementary rather than competing: a token–event graph is cheaper and suits an analysis of history, while a deduplicated states graph suits an analysis of the reachable state space. We compare against two implementations in Section 8, both of which compute the same object we do: the Wolfram/Multicomputation paclet’s `MultiwaySystem` [11]¹, and this project’s own reference implementation (`reference/MultiwayReference.wl`), a direct transcription of the definition used as the correctness oracle throughout this work. The two answer different questions. `MultiwaySystem` is the implementation a user would otherwise run, but every property it exposes is graph-valued, so its time necessarily includes building Wolfram `Graph` objects and no subtraction can remove that from the outside. The reference returns data, so its time is the closest Wolfram-side analogue of the engine’s own native measurement. SetReplace computes a different object, so a wall-time comparison against it would compare two different computations.

Subgraph matching on hypergraphs: HG-

Match [12] matches by hyperedge rather than vertex by vertex, seeding a join from one pattern hyperedge and extending it one hyperedge at a time, and partitions candidate hyperedges by a label-derived signature to restrict enumeration. HGMatch also organizes execution as a dataflow with a task scheduler performing fine-grained dynamic work stealing. Our matcher adopts all three. The join is organized as a seeding scan followed by expansions that each bind one further pattern edge (Section 4.3), and because hyperedges here carry no labels, the signature that partitions candidates is instead the vertex-repetition pattern of an edge (Definition 4.1), which plays the same filtering role for unlabelled data. Where an expansion has a variable already bound, candidates come from the vertex index rather than the signature bucket; on the device this replaced a global signature walk and reduced match-kernel time on one workload from 944 ms to 4 ms.

GPU graph algorithms: Recent work has explored GPU acceleration for subgraph matching [13], graph mining [14], and general graph algorithms [15]. The warp-centric organization of our match kernel—one block per (state, rule) pair, threads striding over seed candidates and each running its own depth-first subtree—follows G2Miner [16]. Our kernel differs from these systems in remaining resident across evolution steps rather than being relaunched per step, because the workload is iterative: each step’s states are the next step’s inputs.

Dynamic graph algorithms: The causal graph’s transitive reduction is maintained as edges arrive rather than recomputed, which places it in the dynamic graph-algorithms setting; Goranci et al. [17] give fully dynamic algorithms for transitive reduction under insertions and deletions, in $\mathcal{O}(m + n \log n)$ amortized time. Our setting admits insertions only, arriving in topological order, so it is a restriction of that problem rather than an instance of it, and we claim no result against it (Section 5.11).

Term rewriting: The theory of term rewriting [18] treats confluence and termination for tree-structured terms, and high-performance implementations [19] optimise similar pattern-matching operations. Hypergraph rewriting differs in that the objects are graph-

¹Distributed through the Wolfram Paclet Repository, <https://resources.wolframcloud.com/PacletRepository/resources/Wolfram/Multicomputation/>; the paclet version each comparison ran against is recorded in that table’s provenance line.

structured, so matching is subgraph matching rather than tree matching, and states must be compared up to isomorphism rather than syntactic equality.

3 Graph Canonicalization

State deduplication determines how large the explored set becomes: without isomorphism detection, every path to a state contributes a separate vertex and the set grows exponentially with depth. Deduplication must be *exact*: two states are the same node if and only if their hypergraphs are isomorphic (vertex relabeling, within-edge order and multiplicity preserved). The exact mode satisfies this requirement using individualization–refinement (IR) alone: it is computed for every state, and its hash serves directly as the deduplication key. Weisfeiler–Leman (WL) is a separate, coarser signature, used by the modes that request a coarser notion of state identity.

3.1 Individualization–Refinement

The exact canonical form is computed by McKay-style individualization–refinement [2], the same family of algorithms underlying nauty and Traces. IR produces a canonical labeling—and, as a byproduct, a set of automorphism generators and vertex/edge orbits—such that two hypergraphs receive identical canonical forms if and only if they are isomorphic.

Definition 3.1 (Canonical Form). *A canonicalization CANON maps each hypergraph to a labeled representative such that $\text{CANON}(\mathcal{H}_1) = \text{CANON}(\mathcal{H}_2)$ iff $\mathcal{H}_1 \cong \mathcal{H}_2$.*

IR proceeds by iterated color refinement to a stable coloring; when refinement leaves a non-singleton cell, it *individualizes* one vertex (assigns it a distinguishing color) and recurses, exploring a search tree of refinements. Automorphisms discovered along the way prune the tree so that symmetric branches are not re-explored, and the lexicographically extremal leaf defines the canonical labeling.

Remark 3.2 (Complexity). *Like nauty, IR is exponential in the worst case and terminates in time polynomial in the state size on instances without large automorphism groups. It gives no polynomial-time guarantee, and we claim none. The worst case is realized on highly symmetric states, which is where the exact mode’s cost is concentrated. No cheaper exact alternative is available to substitute for it: Proposition 3.3 bounds what a Weisfeiler–Leman pre-filter can remove, and Section 8 reports the two implemented alternatives, one measured slower and one that never fires.*

The automorphism generators and edge orbits produced by IR are reused elsewhere in the engine: they define the canonical edge-orbit equivalence used for principled event canonicalization and for exact reconstruction of causal and branchial multiplicities.

Two bounds, and only one of them is evident when it is exceeded. The search is bounded twice: by the individualization depth it may descend to, and by the number of automorphism generators it may record. Exceeding either is reported to the caller, which raises the bound and re-runs, so neither is a silent truncation. The two are not symmetric in what a truncation would cost. A depth bound reached before a discrete coloring means no canonical form was produced at all, and the failure is evident. A generator table filled to capacity, by contrast, yields a canonical form that is correct: the hash is exact regardless of how many generators were recorded. What degrades is the orbit partition, which is obtained by fusing edges over the generators found—fewer generators fuse fewer edges, so the orbits come out too *fine*. Since the orbits are how the quotient reconstruction identifies instances (Section 5.10), a short generator table produces a wrong identification rather than a slow run, and it produces one that looks like an ordinary answer. This is why the generator bound escalates on the same footing as the depth bound rather than being treated as a performance parameter. More generators cannot repair a depth failure, so escalation stops at the first depth report.

The canonical labeling of a symmetric state is a coset. When the automorphism group is non-trivial, IR does not determine a single labeling but a coset of them, and the algorithm’s within-cell tie-break selects one representative. That tie-break reads the vertex numbering it was handed. Two conventions for numbering a state’s vertices therefore select two different representatives, each internally consistent and both agreeing on the canonical *hash*, while disagreeing about which edge receives which rank. Ranks are what identify consumed and produced edges across a rewrite, so the convention has to be fixed once and shared by every caller rather than chosen locally: measured on the equivalence probe’s 4063-state corpus, encounter-order numbering and sorted-unique numbering disagree on 103 states, all of them states with nontrivial automorphisms.

3.2 The Weisfeiler–Leman Signature

The engine also computes a 1-dimensional Weisfeiler–Leman (1-WL) color-refinement hash [3, 9]. WL is a sound but incomplete invariant: isomorphic states always receive the same WL hash, while certain non-isomorphic states — regular graphs with identical local structure, for instance — collide. It is the identity used by the modes that request a coarser notion of state equality. It is not used by the exact mode, for the reason established in Proposition 3.3.

Proposition 3.3 (Ceiling on a WL pre-filter). *Let a run produce R raw states falling into C isomorphism classes. Consider deduplication that computes the WL hash of each raw state, and invokes IR only when that hash has been seen before. Then the number of IR invocations is at least $R - C$, so the fraction of IR calls*

avoided is at most C/R , and one WL pass is paid on all R states.

Proof. WL is isomorphism-invariant, so states in the same class share a WL hash and the number of distinct WL hashes is at most C . A state whose WL hash is already present cannot be resolved by WL: equal hashes are consistent both with isomorphism and with a WL collision, so IR is invoked. At most one state per distinct WL hash finds its hash absent, hence at most C states skip IR and at least $R - C$ invoke it. $\square$

The bound is only useful with C/R in hand, and that ratio is a property of the rule and the depth. Measured across the rule-type corpus, reported per case beside the event count, since the two must be read together (Table 1, plotted in Figure 2):

Table 1: **Canonical-class to raw-state ratio, per corpus workload.** C/R is the fraction of raw states that are distinct up to isomorphism, reported beside the event count because the two must be read together: a small ratio on a small run says less than the same ratio on a large one. This is the quantity Proposition 3.3 bounds the achievable saving by. Source: `tools/cost_matrix.cpp`.

Case	Rule type	Events	C/R	C/R at lower depth
cycle4-automorphic	automorphism	68,184	0%	12%
star4-automorphic	automorphism	68,184	0.3%	20%
disconnected-lhs	disconnected	126	0.8%	14.3%
binary-growth	productive/arity2	873	4.1%	75%
multi-rule	multi-rule	237	15.1%	83.3%
triangle	productive/mixed-conn	124	28.8%	60%
arity3-growth	arity3	153	42.2%	100%
self-loop	self-loop/arity2	1	100%	100%

For every rule that continues to rewrite, C/R decreases with depth. Reaching the same state along several distinct histories is characteristic of multiway evolution, and each such arrival produces a repeated WL hash that IR must resolve. The cases that retain a high ratio are those reaching a fixed point after one or two steps: **self-loop** has $C/R = 100\%$ over two raw states and one event. The fraction a filter could elide is therefore largest on the workloads with the least work to elide.

Two options require automorphism data: quotient exploration, and the event canonicalization conventions of Section 5.15. Both obtain the edge orbits from the same IR pass, so no second canonicalization is performed.

Algorithm 1 Weisfeiler–Leman Hash

```

1: procedure WLHASH( $\mathcal{H}$ )
2:   Initialize  $c_v^{(0)} \leftarrow \deg(v)$  for all  $v$ 
3:    $i \leftarrow 0$ 
4:   repeat
5:      $i \leftarrow i + 1$ 
6:     for each  $v \in V$  do
7:        $c_v^{(i)} \leftarrow \text{HASH}(c_v^{(i-1)}, \text{multiset}\{c_u^{(i-1)} : u \in N(v)\})$ 
8:   until  $\{c_v^{(i)}\} = \{c_v^{(i-1)}\}$   $\triangleright$  partition stable
9:   return HASH(sorted( $\{c_v^{(i)} : v \in V\}$ ))
10: end procedure

```

Proposition 3.4 (WL Complexity). *Algorithm 1 gives the refinement. 1-WL requires $\mathcal{O}(k \cdot |E|)$ time per refinement round for colours of size k , with at most $|V|$ rounds, so $\mathcal{O}(|V| \cdot k \cdot |E|)$ overall (Appendix A).*

This polynomial bound is why WL remains available for the coarser modes. It is not an alternative canonicalizer: where exactness is required, IR (Section 3.1) is the only identity used.

3.3 State Canonicalization Modes

Rather than exposing a choice between competing hashing algorithms, the engine exposes a single *state canonicalization mode*, NONE, AUTOMATIC or FULL, that controls how aggressively isomorphic states are merged. NONE performs no deduplication and keeps every raw state distinct. AUTOMATIC deduplicates by a content-ordered hash of the edge tuples as given. This hash is not an isomorphism invariant: it identifies two states when their edge tuples are literally equal, and separates isomorphic states whose vertices carry different identifiers. FULL is the exact mode used for all correctness-critical results: the IR canonical hash is computed for every state and is itself the deduplication key, so two states are identified if and only if they are isomorphic.

Remark 3.5 (Incremental hashing). *Child states differ from their parents by only the consumed and produced edges, and the engine passes that per-event delta to the canonicalizer. The hash is nonetheless computed from the state’s complete edge set in every mode. No incremental-hashing result is reported here, and none of the measurements in Section 8 depends on one.*

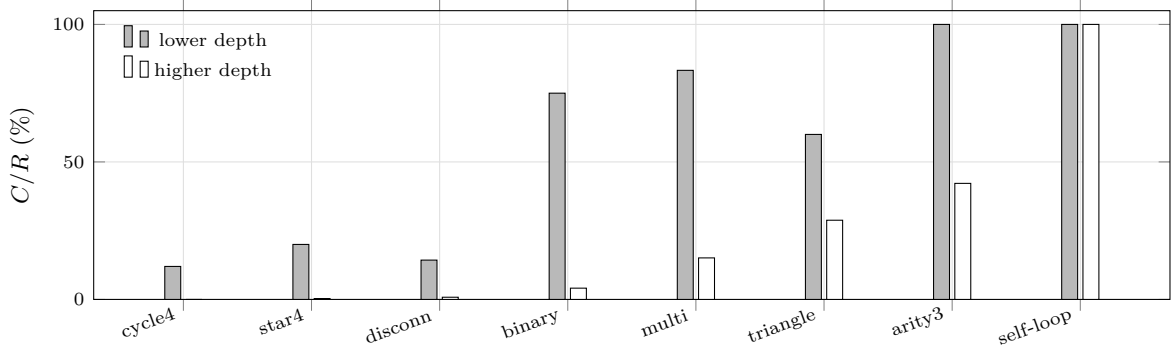


Figure 2: C/R , the ceiling of Proposition 3.3 on the fraction of IR invocations a Weisfeiler–Leman pre-filter could elide, for each corpus case at two depths. The bar pairs are two depth samples per rule and not a continuum, so the figure shows direction rather than a rate. Every case that continues to rewrite decreases. A case that does not decrease is one whose two samples produced the same class-to-raw ratio, which happens where the rule reaches a fixed point before the higher depth—**self-loop** does so after one event, and its ratio is 100% at both. Values as in Table 1.

4 Pattern Matching

Pattern matching—finding all instances of a rule’s left-hand side in a hypergraph—is the first phase of every expansion. On the Wolfram rule, whose states carry at most fourteen edges at depth six, the instruction profile of Table 18 attributes 5.6% of executed instructions to matching, its join and the rewrite together; canonicalization (22.9%) and the quotient reconstruction (23.5%) are the largest components there. Multi-level indexing restricts candidate enumeration.

4.1 Edge Signatures

Definition 4.1 (Edge Signature). *The signature of hyperedge $e = (v_1, \dots, v_k)$ is the pattern $\sigma(e) = (s_1, \dots, s_k)$ where $s_i = s_j$ iff $v_i = v_j$.*

For example, edge $(3, 3, 7)$ has signature $(0, 0, 1)$, capturing the repetition pattern without specific vertex labels. Pattern edges can match data edges only with compatible signatures. For pattern signature σ_p with d distinct variables, the number of compatible data signatures is the Bell number B_d [20]; Table 2 gives the first few. The host tests each candidate edge’s signature against the pattern’s directly; the device precomputes the compatible set per pattern edge (Section 6).

4.2 Candidate Edges by Ancestry

A state is a view into the shared edge pool (Section 5.2), and its matching asks two questions of that view: which of its edges contain a given vertex, once a pattern variable is bound, and which of its edges carry a compatible signature, for the seed. Both are answered from the state’s ancestry. Every edge of a state was brought by exactly one state on its chain of parent states: the root brought all of its edges, and each derived state brought the edges its event produced. Each state indexes its own contribution by vertex once, when it is created, so

the edges of S that contain v are

$$\bigcup_{A \in \text{chain}(S)} \{e \in \text{Contrib}(A)[v] : e \in S\}, \quad (1)$$

where $\text{Contrib}(A)[v]$ is A ’s sorted incidence list for v and membership $e \in S$ excludes an edge that a later event on the chain consumed. A query costs the chain’s length times one search of a small sorted array, nearly every edge it tests is in the state, the index of a derived state is the size of its event’s right-hand side, and nothing is maintained per edge when an edge is created. A global index over the whole evolution’s edges, filtered per state, answers the same question at a cost proportional to every edge ever created at v in any state, of which a state at depth seven holds under two percent.

4.3 Matching Algorithm

Pattern matching proceeds through the task-based system of Algorithm 2:

Algorithm 2 Task-Based Pattern Matching

```

1: procedure FINDMATCHES( $L, \mathcal{H}$ )
2:    $e_1, \dots, e_k \leftarrow \text{MATCHORDER}(L)$   ▷ connected:
   no step is a product
3:   Enqueue SCANTASK( $e_1, \emptyset$ )
4:   while task queue non-empty do
5:      $\tau \leftarrow \text{DEQUEUE}()$ 
6:     if  $\tau$  is SCANTASK( $e_i, \phi$ ) then
7:       for each  $e \in S$  with  $\sigma(e)$  compatible
       with  $\sigma(e_i)$  do
8:         if COMPATIBLE( $e, e_i, \phi$ ) then
9:           Enqueue EXPANDTASK( $e_{i+1}, \phi'$ )
10:        else if  $\tau$  is EXPANDTASK( $e_i, \phi$ ) with  $i > k$ 
11:        then
12:          yield complete match  $\phi$ 
13:        end if
14:      end while
15:    end procedure

```

Table 2: Bell numbers for signature compatibility

d	1	2	3	4	5	6	7	8
B_d	1	2	5	15	52	203	877	4140

The order $e_1, \dots, e_k$ is fixed per rule when the rule is added, and it is required to be connected—every edge after the first shares a variable with one already bound—so no step of the join is a cartesian product. Two host paths execute that order: a synchronous depth-first join, which runs an entire search inside one task and spawns a job only for a completed match, and the task-spawning form above, which spawns a task per candidate and is what exposes the search itself to work stealing. They are not two joins. The recursion, the edge-injectivity check, the binding and unwind, and the rule choosing which pattern position is bound next are one shared body, which is also the body the device runs; the two paths differ in who executes the resulting steps and when.

4.4 Position Against the Worst-Case-Optimal Bound

Treating a left-hand side as a conjunctive query, the AGM bound [21] gives the maximum number of results a query can have on a database of a given size, and a join is worst-case optimal if its running time matches that bound up to logarithmic factors [22]. Our join is edge-at-a-time: it binds one pattern hyperedge per expansion, in the connectivity order of Section 4.3.

For left-hand sides of one or two hyperedges, an edge-at-a-time join meets the bound, and every rule in the ORACLE corpus of Section 8 is of that form; the left-hand-side sweep of Section 8.8 deliberately goes past it, to three and four edges. It does not meet the bound in general. On a cyclic left-hand side of three hyperedges the AGM bound is $N^{1.5}$ for a state of N edges, while an edge-at-a-time join performs $\Omega(N^2)$ work: two of the three edges can be bound before the third constrains anything, and the intermediate result is already quadratic. No choice of join order removes this, since the difficulty is the granularity of binding rather than its sequence. Closing it requires enumerating variable-at-a-time in the manner of a worst-case optimal join, which in turn requires indices supporting ordered seek on a bound variable rather than the incidence lists described above.

We therefore state the position rather than claim the property: the join is worst-case optimal on the rule shapes measured here, and is not worst-case optimal in general. The static analysis of Section 5.6 computes, per rule, whether its left-hand side is acyclic and what its integral edge cover is, so a rule set for which this gap matters is identifiable before it is run.

4.5 Match Forwarding

When state s' is derived from s , a match valid in s remains valid in s' unless it used an edge the rewrite

removed. The engine forwards such matches from parent to child rather than rediscovering them:

Proposition 4.2 (Match Forwarding). *Let $\mathcal{M}$ be a match in state s , and let s' be derived from s by applying rule r at match $\mathcal{M}'$. Then $\mathcal{M}$ remains valid in s' iff $\text{edges}(\mathcal{M}) \cap \text{edges}(\mathcal{M}') = \emptyset$.*

Forwarding therefore avoids re-running the join for every match a child inherits unchanged from its parent.

Delivering it under concurrency requires two mechanisms rather than one. A match discovered in a state is *pushed* to the children that state already has, which covers every child registered at the time of discovery. A state created later cannot receive a push that has already happened, so on creation it *pulls*, walking the chain of ancestors and collecting each one’s matches that its own rewrite did not invalidate. Neither half suffices alone: without the push, every state would repeat its ancestors’ work on creation; without the pull, a match discovered before a child existed would never reach it.

The two halves interact through an ordering requirement. The pull’s ancestor walk treats a missing parent link as having reached a root and stops there. A child must therefore publish its parent link *before* it becomes visible in its parent’s child list: once it is visible it may receive a forwarded match, that match may create a grandchild on another worker, and that grandchild’s pull walks upward through this child. Were the link published after visibility, such a walk could find it absent, stop early, and miss every match held only in higher ancestors—without any indication, because pulled matches are not re-stored in the descendant and pushes happen once at discovery, so nothing later repairs the omission. Publishing the link first makes an absent link mean *root* for every walk that can reach the state.

Forwarded matches are accumulated per parent and delivered to the child together rather than one at a time, and whether forwarding is applied at all is decided per rule set by the static analysis of Section 5.6; Section 8 reports the ablation.

4.6 Delta Matching

Forwarding supplies a child with the matches it inherits unchanged from its parent. Its complement is forced: a match present in the child but not in the parent must use at least one produced edge, because a match built entirely from edges that survived the rewrite was already a match in the parent. The child’s own scan is therefore restricted to matches anchored on a produced edge, which is a small set—one rewrite produces the right-hand side’s edges, independently of how large the state is.

The two mechanisms partition the child’s matches rather than overlapping: every match either touches a produced edge or does not, forwarding covers the second class and the delta scan the first. What does overlap is the delta scan with itself. A match touching k produced edges is discovered k times, once with each of them as the anchor, so the results are deduplicated on content rather than on the anchor that found them; duplicates here are routine rather than exceptional, so the deduplicating lookup precedes the allocation of a match record rather than following it.

Because the partition is an argument rather than a measurement, the engine carries a validation mode. With the mode enabled, every state’s matches are additionally computed by an unrestricted scan and compared against the union of what forwarding and the delta scan supplied, and a missing match is attributed to the obligation that owed it: to forwarding if it uses only surviving edges, to the delta scan if it touches a produced edge. The two have different causes and different repairs, so the attribution is recorded separately rather than as a single mismatch count. The validator also counts its own executions, since a mismatch count of zero is evidence only when the number of comparisons performed is not.

5 System Architecture

The architecture follows from two commitments that constrain every structure below. The first is that a rule—an ordering, an identity, a hash, a predicate—has exactly one implementation, compiled for both the host and the device, so the two targets cannot diverge on what a state *is*. The second is that no thread ever blocks another: every shared structure is lock-free, and where a protocol needs agreement it reaches it by a single compare-exchange whose loser adopts the winner’s answer rather than waiting. The subsections take these in turn—the shared core, the storage the states live in, the lock-free structures, the allocator that keeps them off the global heap, the memory model that makes their orderings checkable, and the static analysis that decides how a rule will be matched before any matching happens.

5.1 One Implementation per Rule, Compiled for Both Targets

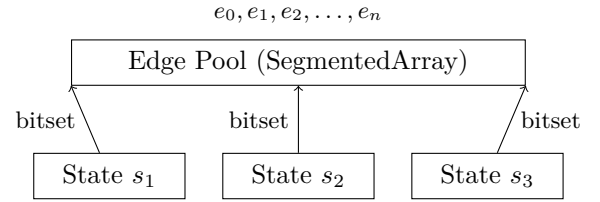
The CPU engine and the CUDA port are separate schedulers over the same semantics, and any rule implemented twice will eventually be implemented differently. Every decision procedure the two share therefore exists once, as an annotated function compiled for host and device from a single source: the join that enumerates matches, the edge-signature rule, the rewrite semantics determining which vertices a produced edge carries, the event identity, the Weisfeiler–Leman hash, the individualization–refinement canonical hash, and the quotient-causal dynamic program and its per-instance replay. Eleven such cores are shared in total; the three

not named here are the matcher’s task core, the sampling draw and the slot rule.

The consequence is that host and device agree by compilation rather than by testing, and a differential test between them can be reserved for the parts that genuinely differ—scheduling, memory placement and queueing—rather than re-checking a rule that exists once. What the arrangement rules out is a rule written twice: two implementations of one decision agree until one of them is edited, so the duplication is the defect and not the disagreement it eventually produces.

5.2 Unified Hypergraph Storage

Our core data structure stores all hypergraph data in a unified edge pool, shown in Figure 3: every state is a bitset view over one shared pool of edges, so a child state that differs from its parent by a few edges stores that difference and not a copy.



States are views (SparseBitset) over shared edges

Figure 3: Unified edge storage with state views

This design eliminates edge duplication across states, since each state is a bitset view over the shared pool rather than an independent copy of its edges. The bitset is chunked and the chunks are shared: a child state initially references its parent’s chunks, and a chunk is copied only when the child first mutates it. A rewrite that touches a few edges therefore allocates a few chunks rather than a copy of the parent’s membership, and chunks are immutable once shared, so concurrent readers of a parent are unaffected by a child’s mutation. Table 5 reports the arena bytes each corpus workload consumes.

5.3 Lock-Free Data Structures

Concurrent evolution requires thread-safe data structures without lock contention. The constructions below follow standard non-blocking techniques [23], in particular compare-and-swap insertion into an open-addressed table [24] and lock-free list linkage [25]:

ConcurrentMap<K, V>: Hash table using atomic compare-and-swap for insertion. Wait-free lookup with lock-free insertion.

ConcurrentKeySet<K>: A set, for membership rather than mapping. Presence and ownership are decided by one compare-exchange from the empty sentinel to the key, so none of the map’s value-publication protocol exists: no locked slot, no settling of a claim met mid-flight. An insert

scans the superseded-table chain for a settled copy of its key before claiming at the head, and a claim that lands in a table which is no longer the head is reported stale and re-driven rather than counted—the two halves of keeping one key from ever being claimed twice. Growth carries each key into the new table and then seals the slot it came from with a second reserved key, so a key reachable through two tables is enumerated once, and a table whose slots are all sealed is skipped rather than probed—without which a retired table is scanned in full on every operation, because sealing removes the empty slot linear probing stops at. Membership queries walk the chain oldest table first, and Section 7’s reader-visibility harness checks that order. The quotient reconstruction’s membership marks use it (Section 5.10).

SegmentedArray $\langle T \rangle$: Dynamically growing array stored in fixed-size segments. Append-only with atomic size tracking.

LockFreeList $\langle T \rangle$: Append-only linked list for index entries. Supports concurrent iteration via snapshot semantics.

Work-stealing deque: A bounded Chase-Lev deque [26]. Each worker owns one, pushing and popping at its own end without a contended atomic in the common case, and stealing from the opposite end of another worker’s deque when idle. Its memory orderings follow Lê et al. [27]. Following HGMatch [12], evolution is expressed as a dataflow over such a scheduler: a scan task seeds a join, expansion tasks extend it, and completed matches are handed to rewriting, with work stealing providing load balance across a frontier whose width is not known in advance.

Job pool: Tasks are drawn from a per-thread slab rather than allocated, so submitting a task performs no call to the global allocator, so the allocation figures of Section 5.4 hold for a workload that submits one task per candidate expansion. Allocation and same-thread release touch only the owning thread’s private free list, with no atomic and no contention. Work stealing means a task allocated on one thread is often released on another, so a foreign release is pushed onto the owner’s separate foreign list by compare-and-swap, and the owner reclaims the whole accumulated batch with one exchange when its private list empties. That shape is free of the ABA problem by construction rather than by a tag: producers only ever prepend, and the single consumer takes the entire list at once, so there is no node for a compare-and-swap to observe unchanged after reuse. Each slot carries a header naming its owning pool, so a release on any thread reaches the pool the slot came from; an object too large for a slot falls back to the general allocator, so correctness does not depend on the slot size being estimated correctly. Pools are never destroyed and are recycled through a fixed array

of claimable slots when a thread exits—an array rather than a stack of free pools, because a pool that is claimed, used and released again is exactly the reuse a stack’s head compare-and-swap cannot distinguish, while an array slot has no pointer to compare.

The shared injector, and what bounds its ABA defence.

Work submitted from outside a worker, and work that overflows a worker’s own deque, goes to one shared double-ended queue. Its state is a single 64-bit word packed as a 32-bit tag and two 16-bit indices, so one compare-exchange commits both ends at once and a pop at each end racing for the last item resolves by one of the two exchanges failing. The three fields share a word because a 128-bit compare-exchange is not lock-free on the supported targets; the indices need 16 bits each for a capacity of 32,768 entries, which leaves 32 bits for the tag. Because both ends move in both directions, the index pair recurs, and the tag lets a compare-exchange commit only if nothing changed since the load.

A 32-bit tag wraps, so the defence is conditional, and the condition is stated here rather than assumed. For a stale item to be claimed twice, the tag must complete a full cycle *and* both indices must return to the values one popper read, all within that popper’s own load-to-exchange window—a handful of instructions, so it requires the thread to be descheduled inside them. The throughput measurement that bounds this is direct: at eight threads the shared queue sustains 8.16×10^6 operations per second, so 2^{32} of them take on the order of 500 seconds of continuous stall while every other thread runs flat out. The same measurement gives the cost of sharing one word: 1.33×10^8 operations per second at one thread against 8.16×10^6 at eight. Per-worker deques are near-exclusive to their owners and do not incur that reduction; the injector is shared and does, so it holds only overflow and external submissions.

Neither queue offers a blocking push or pop. For a worker this is not a preference but a requirement: a worker parked inside a push cannot pop, so if every worker parked there the queues would never drain again—and since expansion is combinatorial and submits from inside running jobs, saturating a local deque and the injector together is reachable rather than hypothetical. A full queue is answered by running the job on the calling thread, which always makes progress.

A worker with no available work does not spin indefinitely. It waits on a 32-bit address through the platform’s own primitive—`futex` on Linux, `WaitOnAddress` on Windows, `os_sync_wait_on_address` on macOS—so a blocked worker consumes no cycles and is woken by the thread that supplies the work. No mutex or condition variable is used on any of these platforms: the engine’s object files reference none, and the only such construction in the tree is a fallback compiled solely for a platform providing none of the three primitives.

One code path for a machine with no threads. The scheduler also runs with no worker threads at all. In that mode none are spawned: submission routes every job to the injector and the wait call drains it inline, in first-in-first-out order, on the thread that called it. Jobs submitted by a running job land behind it and execute in submission order, so a run in this mode is deterministic by construction rather than by the argument of Section 5.8. This is the single-threaded execution mode, and it is also the WebAssembly target, where spawning threads is not available. Nothing above the scheduler distinguishes the two: the engine is compiled once and the work the workers would have done happens on the thread that waits.

5.4 Allocation

Evolution allocates from per-worker arenas rather than from the global allocator. Each worker holds a bump cursor into blocks it owns, so allocation on the matching and rewriting paths is a pointer increment with no synchronization, and memory is reclaimed in bulk by resetting the arena rather than by freeing individual objects. Calls to the global allocator are consequently confined to acquiring those blocks: Table 5 reports 455 to 475 such calls for an entire run, which on the largest workloads measured is 0.007 calls per event.

5.5 Memory Model

A new state is published to readers by a release store paired with an acquire load, so a reader that observes the state also observes the edge set written before it. A producer appends the new edges, inserts their ids into the state’s edge set, issues a release fence, and only then appends the state; a consumer that finds the state through the canonical map’s acquire load issues an acquire fence before reading the edge set. The release-acquire pair orders every edge write before the state’s publication, so no reader of a published state can observe an edge set still being filled.

5.6 Static Rule Analysis

Some properties of a rule set are decidable from the patterns alone, before any evolution runs, at the cost of a bounded scan over two finite patterns. The engine computes them when a rule is added and uses them to select strategy.

Per rule it records the edge-count difference between the two sides, which predicts whether states grow and therefore how canonicalization cost will develop; the number of variables the right-hand side introduces, since a rule creating none operates within a bounded vertex set; and the left-hand side’s arities. Treating the left-hand side as a conjunctive query, it computes whether that query is acyclic—which determines whether a join order exists that avoids intermediate blow-up—and an integral edge cover, which bounds the number of matches a state of N edges can yield by N^p . That bound is tight exactly when the query is

acyclic; where it is not, the recorded value is an upper bound and is used only as one.

Every predicate is sound in one direction only, and each is stated with its direction. The branching test over-approximates: a negative result means branching is structurally impossible, while a positive result means only that it has not been ruled out. An adaptation may act on the negative and must not act on the positive. The engine does exactly that with the branchial relation: if no two matches can share a consumed edge then no state has a branchial pair under any initial condition, so the relation is provably empty and is not built. The request is not overridden by this—what the caller asked for is the empty relation and the empty relation is what it receives; only the work of discovering it is removed. Nothing here concerns termination, finiteness of the reachable set, or global confluence, all of which are undecidable for a rewriting system of this power.

5.7 Behavior at Configured Limits

The engine’s containers are sized from configuration, and a workload may exceed them. Reaching a configured ceiling is treated as a distinct outcome from a programmer error: the run returns the portion of the evolution that fits, together with a warning naming the limit that was reached, rather than raising an exception and discarding the work. Both backends behave identically in this respect. Options that a build does not implement are reported the same way, so a caller who requests one receives the result computed without it and a statement to that effect, rather than silently receiving a different computation from the one requested.

5.8 Determinism Contract

The engine’s observable output—the state set, the event set, and the causal and branchial relations—is a deterministic function of the rules, the initial state, the step count and the options. It does not depend on the number of worker threads, on the order in which the scheduler happens to execute tasks, or on whether quotient exploration is enabled. Identifiers assigned to states and events are allocated in arrival order and therefore do vary between runs; every observable defined above is stated over canonical identities and is invariant. This property is checked by a determinism gate that fingerprints the expanded state set, the (input, output, rule) transition multiset, and the causal and branchial relations across thread counts and seeds, and on a divergence reports which component moved and under which configuration. Reporting the components separately makes the gate sensitive: the causal relation can move on its own, by a single edge, while the state set, the transition multiset and the branchial relation stay identical, so a gate that compared one aggregate fingerprint would not distinguish that from agreement.

The capping options are outside it, for the reason

that shapes the sampling design: MAXSTATESPER-STEP admits states until a per-step counter reaches its bound, so which states are admitted follows the order they arrive in and varies with the worker count. An exact cap that was also schedule-independent would have to hold every candidate at that depth until the depth settled, which is the barrier Section 5.13 declines: only a population that completes locally can be finalised without one, and a per-step population does not. What a capped run guarantees is therefore the bound, at every worker count, and not which states meet it. The per-parent cap has a locally-completing population and could be made schedule-independent on the same terms as the transition spine; it is not today.

5.9 Depth as a Shortest Path, and Termination

A run is bounded by a number of steps, and the bound is applied to the *shortest* path length from an initial state to a canonical state rather than to the length of the path along which the state first happened to be reached. The distinction has no content sequentially, where the search reaches states in order of increasing depth. It has content under concurrency: a state can be reached first along a long path and only later along a short one, so a budget applied to arrival depth would admit or exclude states according to which path won a race, and the set of states expanded would vary with the thread count.

The engine therefore *relaxes*. When a state is reached along a path shorter than its recorded depth, the recorded depth is lowered and the reduction is propagated into the state’s children, which may in turn be lowered, and so on. The cascade terminates because depth strictly decreases on each accepted relaxation and is bounded below by zero. Matching is not repeated by it: the claim that admits a canonical state to expansion is a separate value from its depth, so a state whose depth falls is matched at most once regardless of how many times it is relaxed.

Relaxation and child registration race, and the race is settled by ordering rather than by exclusion. Registering a child publishes it into its parent’s child list and then reads the parent’s depth; relaxing a parent lowers the parent’s depth and then scans the parent’s child list. Sequentially consistent fences on both sides exclude the one outcome that would strand a child at a stale depth—the scan missing the child *and* the read missing the lowered depth—so a child always learns its parent’s true current minimum, whichever side of the race it is on. A state that lies past the budget is retained as a frontier state for a continuation to resume from rather than discarded, and its expansion claim is deliberately left untaken, because a shorter path found later in the same run must still be able to relax it into budget, and a state that has already been claimed never would.

Budget consumption is claimed by compare-and-swap rather than by an unconditional increment fol-

lowed by a rollback on overshoot. The two differ in what other threads observe: a rollback publishes a count above the limit for as long as it takes to undo, and the readers that prune work against the budget read exactly that count, so N concurrent claimants would make each other’s in-budget work appear out of budget and which work was pruned would depend on the interleaving. A compare-and-swap claim never publishes a count above the limit, so the budget prunes the same work under every schedule.

Termination is detected without a barrier. A depth is settled when it holds no live work and the depth below it has already settled; settling one depth cascades to the depth above, which may have been waiting only on it. The argument that these two conditions suffice is that a task at depth d submits work only at depths greater than d , so once $d - 1$ is settled nothing can place work at d again. No thread waits on any other, and the check is a test of two values rather than a rendezvous.

5.10 Quotient Exploration

Under full expansion, every raw state that is produced is itself expanded, so a state reachable along k distinct histories is matched against k times. Because the canonical hash already identifies these as one state, that work is redundant for the purpose of discovering new states.

Quotient exploration retains the events but not the repeated expansion: a state found equivalent to one already seen is still created, and its incoming event is still recorded, so the event set and the causal and branchial relations are unchanged; what is suppressed is the spawning of match tasks from that state. Exploration therefore proceeds from one representative of each isomorphism class, at the shallowest depth at which the class was reached. Table 7 reports events produced, causal edges and wall time for the two modes at increasing depth.

The two modes agree on the canonical state set by construction. That they also agree on the causal and branchial relations is less immediate, because those relations are defined over raw events and quotient exploration expands fewer states.

The reconstruction is a forward tally. Under an exact identity the canonical states graph is cyclic, since a canonical state recurs at several path lengths; its unrolling by depth is acyclic, and the multiplicity of each (canonical state, depth) pair satisfies

$$\begin{aligned} \text{mult}[s_0, 0] &= 1, \\ \text{mult}[c, k+1] &+= \text{mult}[s, k] \cdot w(s \rightarrow c), \end{aligned}$$

summed over the transitions into c , with the raw event count recovered as $\sum \text{mult}$. Every quantity propagates forward over the unrolled DAG, so a *count* is recovered by summation rather than by re-exploring, and that part is polynomial in the number of (canonical state, depth) pairs.

Recovering the raw *observables* is a different matter, and the distinction is the one that decides what

quotient exploration costs. Mapping a multiplicity back to individual instances requires distinguishing the edge roles within a class, which the edge orbits from the same IR pass provide (Section 3.1); no additional canonicalization is performed. But the instances are then materialised one by one, and there are as many of them as the full system holds. On a two-rule set whose canonical state count grows linearly with depth—three, four, five, six, seven, eight states at depths two through seven—the raw system reached 146,599 states by depth six, and the reconstruction’s cost tracked that rather than the canonical count. Quotient exploration visited 49 raw states there against full capture’s 146,599 and still ran only 65 times faster: a large constant factor, not the asymptotic saving the state counts suggest.

Quotient exploration and raw reconstruction are separate requests, and the implementation keeps them separate. The reconstruction runs only when the causal relation, the branchial relation or the raw event set is among the requested outputs, so a run that asks for canonical states alone performs neither the materialisation nor the growth that accompanies it. The reconstruction is also not a single-threaded pass—every worker drives it, and its total cycle count grows with the thread count, which Section 8.16 records among the measured limits.

5.11 Transitive Reduction of the Causal Graph

The causal graph accumulates an edge for each dependency between events, including those implied by longer paths. The transitive reduction removes an edge $p \rightarrow c$ whenever a directed path $p \rightsquigarrow c$ of length greater than one already exists; on an acyclic graph this reduction is unique.

Maintaining a transitive reduction under edge insertion is an instance of the dynamic transitive-reduction problem. Goranci et al. [17] give the first fully dynamic algorithms for maintaining a transitive reduction of a general directed graph under both insertions and deletions, achieving $\mathcal{O}(m + n \log n)$ amortized update time, and $\mathcal{O}(m + n^{1.585})$ worst case using fast matrix multiplication. The setting here is strictly easier in two respects: causal edges are only ever inserted, never deleted, and they arrive in an order consistent with a topological order of the causal DAG, since an event’s producers are complete before the event exists. We therefore make no claim of novelty for the procedure below, and no claim of optimality; it is the specialisation the easier setting permits, and a fully dynamic reduction would be required if deletions were admitted.

Why the reduction is maintained rather than computed. Computing the reduction as a pass over the finished graph would require the graph to be finished, which conflicts with two properties the engine offers. A session (Section 5.14) may be queried between steps, so there is no point at which the causal graph is known to be complete; and the reduction is what most consumers of the causal graph actually want, so mate-

rialising the full relation first and reducing afterwards would allocate the transitive closure that the reduction exists to avoid. The reduction is therefore maintained as an invariant: after every insertion, the retained edge set is the transitive reduction of the causal DAG seen so far.

Procedure. Let $e = (p, c)$ be a new causal edge, meaning event p produced an edge that event c consumed. Insertion asks whether c is already reachable from p through the retained edges; if it is, e is implied and is not retained, and if it is not, e is retained. There is no removal step; the arrival discipline makes one unnecessary. A consumer’s incoming edges are registered by that consumer’s own worker in descending producer-event order, and an ancestor’s id is strictly smaller than its descendant’s. A retained (p', c) could become implied only through a path $p' \rightarrow \dots \rightarrow x \rightarrow c$; if (x, c) was registered before (p', c) then the path was already visible to the arrival query, which drops (p', c) on the spot, and if after, then $x < p'$ in the descending order while reachability from p' requires $p' < x$ —a contradiction. An edge accepted at arrival therefore stays non-implied forever, and deciding each edge once maintains the exact reduction. The descending order also makes the queries cheap: the closure reachable from nearer producers is established before farther producers are tested against it, and a farther producer is usually eliminated by a query that terminates early. Because events are numbered in creation order and an event’s producers necessarily precede it, the order is available without sorting.

What the arrival order establishes. Two properties of this setting make the problem strictly easier than the fully dynamic one. Causal edges are only inserted, never deleted, so no retained edge can become non-implied later, and an edge dropped as implied never has to be reconsidered. And edges arrive in an order consistent with a topological order of the causal DAG: an event exists only once its producers do, so when (p, c) arrives, every edge into p has already been processed and the closure below p is final. A fully dynamic algorithm must maintain reachability under deletions, and that requirement forces the $\mathcal{O}(m + n \log n)$ amortized bound of Goranci et al.; neither of the two properties above is available to it.

Order independence. The retained edge set is required to be a function of the causal DAG alone and not of the order in which edges were inserted. This is not automatic: a concurrent evolution inserts causal edges from several workers, and a reduction that depended on interleaving would produce different graphs on different runs of the same input. The requirement is the same schedule-independence property as Section 5.8 and is checked by the same gate, which compares the reduced edge set across thread counts and seeds. It is secured differently on the two paths that serve the relation. Full capture holds the arrival discipline of the procedure above—each incoming edge registered by its consumer’s

own worker, in descending producer order, against ancestors whose own registration is complete—and under it the drop-at-arrival decision is exact whatever the interleaving of consumers. The quotient reconstruction emits between canonical event ids, which are not monotonic under recurrence, so no arrival discipline exists there; it stores the relation unreduced and reduces on read, where the uniqueness of a finite DAG’s transitive reduction makes the answer a function of the stored set alone.

The reduction applies only to the causal graph. States, events and the branchial relation are never reduced: branchial edges connect states at the same step and carry no transitivity to remove.

5.12 Pruning and Sparse Exploration

Exhaustive evolution is not always the object of interest, and the state count grows too quickly for it to be available at large depth. The engine bounds an exploration by four independent mechanisms, which compose:

- **Hard caps.** The total state count, the total event count, the states admitted per step (`MAXSTATESPERSTEP`) and the successors kept per parent (`MAXSUCCESSORSTATESPERPARENT`) each bound the exploration through an atomic counter, so the *count* retained is exact at every worker count. Which states fill a cap follows arrival order, which is the one deliberate exception to the determinism contract of Section 5.8, made for the reason given there: an exact cap that was also schedule-independent would require the depth barrier the design declines. A run that reaches a cap returns the bounded evolution with a warning naming the cap, the same contract as a configured container limit (Section 5.7).
- **Probabilistic thinning.** A retention probability (`EXPLORATIONPROBABILITY`) and per-rule transition rates keep a random subset of transitions, keyed on canonical identity so the sample is reproducible under a seed. This is the sparse exploration mode; its design is the subject of the next subsection.
- **Structural pruning.** Quotient exploration (Section 5.10) expands one representative per isomorphism class and reconstructs the raw relations exactly, so the pruned population is not an approximation of the full one but a compression of it.
- **Steering.** A session (Section 5.14) exposes the frontier after each budgeted step, and a continuation may narrow the frontier to the states of interest; steering narrows what runs next and never discards what has been computed.

The mechanisms differ in what they preserve. Caps and steering keep exact computation over a subset

chosen by count or by the caller; thinning keeps a statistically faithful sample; quotienting keeps everything, expanding less. The depth budget bounds all four, and the reported frontier (Section 5.14) records where any of them stopped.

5.13 Sampled Exploration

The sparse mode applies sampling at the transition rather than at the state: each candidate transition is retained with a fixed probability, so the number of children a state contributes remains a random variable rather than being truncated to a fixed count. Truncating to a fixed count per state would collapse the offspring distribution to a point mass and remove the property the sample exists to estimate.

Per-rule weights multiply that retention probability, so a rule may be explored fully while another is suppressed entirely, and the two controls compose rather than one overriding the other. A seed makes a sampled run reproducible; sampling decisions are keyed on canonical identity, so the same seed yields the same sample at any thread count, consistent with Section 5.8.

Choosing the rate. A fixed retention probability applied at every depth is not a neutral choice, because the expected number of surviving children per state compounds with depth. On the workloads measured, retention of 1/8 and 1/4 extinguish the frontier within the first step, while 1/2 leaves the population growing; the useful range between extinction and no thinning is narrow and depends on the rule’s branching. The rate is therefore permitted to vary with depth, so that a sample can be thinned without the frontier dying before it reaches the depth of interest.

Coverage is limited by reachability, not by survival. A per-transition coin does not recover the exhaustive system as seeds are accumulated, and the reason is structural rather than a matter of the rate. A transition can only be sampled if its source state was created, and its source state was created only if some earlier coin fired, so what a run misses is not a scattering of individual transitions but every subtree below a transition that was dropped. Two attempts to improve coverage by adding independent randomness—an extra coin per arrival depth, and an extra coin per arriving ancestor class—left the union-over-seeds recovery curve unchanged when measured, which is the signature of a limit on reachability rather than on per-transition survival.

The control that does move that curve keeps a connected skeleton. A cap of k transitions per (state, rule) pair retains at most k of a state’s own transitions for each rule, selecting them by a deterministic pseudorandom rank of the canonical transition key mixed with the run’s seed. Every state that is reached therefore keeps at least one outgoing transition, so the sample is a spanning substructure rather than a thinned scattering, and a different seed keeps a different skeleton—so

the union over seeds grows in reachability, which is the quantity that was limiting.

The selection is made when the state’s matching completes rather than as matches arrive, because choosing k of M requires all M , and M is known only at that point. Selecting by arrival order would instead make the retained set depend on the schedule. On the same ground the cap ranges over a state’s own matches only and not over matches forwarded to it from an ancestor: a forwarded match races the completion, so a cap that counted them would retain a different set at a different worker count. Forwarded matches are instead thinned individually, on the same terms as discovered ones, at the point they arrive—without which the sampled subgraph would depend on whether forwarding was enabled.

Finally, the per-state and per-transition retention controls are one knob under full capture and two only under quotient exploration. Under full capture raw states stand in bijection with the events that create them, so one coin per new state and one coin per transition are the same coin; under quotient exploration a canonical state has many incoming transitions and the two separate.

Sampling and synchronization. Sampling constrains parallelism through the order in which decisions become final. A sample cannot be acted on before its population is complete: a rewrite cannot be undone, so a match that has been applied cannot later be evicted by a match that arrives afterwards. Any reservoir must therefore be finalised before its members are rewritten, and finalising requires knowing that the stream of candidates has ended. A scheme that samples from all matches produced in an evolution step consequently needs every worker to finish that step before any rewriting begins—a global barrier, in an engine that otherwise has none.

The question is therefore not whether to have a barrier but how local the completion condition can be made. The engine requires a sampling population to be one that *completes locally*: its completion must not depend on knowing anything about work outside the object it is keyed on. A population keyed on a single (state, rule) pair satisfies this, because the engine can tell when that pair has no matches outstanding without consulting any other state; a population keyed on an evolution step does not.

Restricting to locally-completing populations removes less than it appears to, because the global population was not delivering a uniform sample either. Under the step-wide scheme each match job received a reservoir of the same capacity regardless of how many matches it saw, so a pair producing a thousand matches contributed no more samples than one producing twenty; the pooled sample was already weighted toward sparse strata before any shuffle, and shuffling a skewed pool does not remove the skew. Uniformity over the union of all matches was not a property that scheme had. Uniformity within a (state, rule) stratum was, and is the property its test checks. Keying

the population on the state therefore preserves what was true and removes the barrier.

5.14 Sessions

Evolution is also exposed as a session rather than only as a single call. A session is opened on a rule set and initial condition, advanced by a requested number of steps, queried for any of the derived structures, and closed. The unexplored frontier is retained between steps, so advancing a session continues the existing exploration instead of recomputing it from the initial state, and a caller may inspect intermediate results before deciding how much further to evolve.

The frontier is a separate structure from the identity map, and it has to be: one bit cannot answer both questions a continuation asks. The deduplication map records that a state has been seen, and a worker reads that same answer as meaning the state has been expanded. Carrying only the map across calls therefore fails in one of two directions depending on who owns it, and both were measured on the same workload—a two-edge to three-edge rule advanced to depth two and then to depth three, against seven states and six events for a single run to depth three. Rebuilding the maps per call gave ten states and nine events: nothing was recognised, so the second call re-derived as new every state it already had. Giving the caller ownership of the maps gave five states and four events: everything was recognised, so nothing was re-expanded and the new depth was never reached. A session therefore carries both the maps and a frontier—the states whose expansion was refused by the step budget rather than by deduplication—which is the distinction the two failures are between. The device implements the same separation as the host, since a continuation that behaved differently on the two targets would not be the same session.

Every session reply reports the frontier, because a caller cannot steer a continuation toward states it cannot name, and a step may name a subset of it: the named states are expanded and every other frontier entry is put back, so steering narrows what runs next, never what remains reachable. Naming a state that is not on the frontier is an error rather than an empty step—a caller steering toward a state the exploration has already passed would otherwise get a silent no-op indistinguishable from a branch with no successors. Retention is also why each frontier entry carries its own depth: after a steered step, entries stranded by different budgets sit on the frontier together, and an entry resumed at a uniform depth would have its subtree truncated by steps it never took. Both devices serve this contract, resolving the selection against the frontier as it was last reported.

5.15 Event Canonicalization

Events can be canonicalized at multiple granularities:

ByEndpointStates: Two applications are the same event when they run between the same pair of

canonical states. $\mathcal{O}(1)$ signature computation.

ByConsumedProducedEdges: The signature additionally carries the canonical ranks of the edges the application consumed and produced. $\mathcal{O}(E)$ signature computation.

DistinctApplications: No signature is computed and every application is its own event. This is the identity the raw causal graph is defined over.

The three refine one another in that order, so the canonical event count is non-decreasing across them; Table 19 reports it per workload. BYCONSUMEDPRODUCEDEDGES requires the edge correspondence between isomorphic states, which is the per-edge canonical rank produced by the same individualization-refinement pass as the state’s hash.

Event identity does not inherit the state-identity mode. Event identity is defined over isomorphism classes of states, and the specification makes it independent of which state-identity mode a run selects. Resolving it through whichever hash the state mode happens to use would break that independence. The Weisfeiler–Leman hash is isomorphism-invariant but strictly coarser than the exact form, so outside the exact state mode more states share a representative, the edge correspondence between an event’s endpoints resolves to a coarser one, and the event identity derived from it coarsens with it. The consequence is measurable rather than theoretical: resolving event identity through the approximate hash reports fewer distinct events on the binary-growth corpus case than the exact resolution does, a difference produced entirely by the state mode leaking into a quantity that is not supposed to depend on it.

The engine therefore resolves event representatives through the exact invariant in *every* state mode whenever event canonicalization is enabled, independently of the hash the state mode uses for state identity. The two are separate values: one decides state identity, the other resolves event representatives. The exact hash is computed only when event canonicalization requires it, so a run with event canonicalization disabled does not compute it and behaves as its state mode specifies. When it is enabled, the same individualization-refinement pass yields both the exact hash the event path resolves through and the per-edge ranks that identify consumed and produced edges, so the guarantee costs one pass per state rather than two per event.

Configuration dependencies are reported, not absorbed. The canonical-class event identity and quotient exploration are both defined over canonical states and their edge orbits, and those orbit tables are produced only by the exact state-canonicalization mode. A run that asks for the first while selecting a weaker state mode is therefore asking for something the configuration does not supply. The engine neither raises the state mode behind the caller’s back nor proceeds as though the request had been met: it builds

the causal graph by the raw-edge rendezvous, which is well defined without orbits, and returns a warning naming the dependency and the setting that satisfies it. The general form of the rule is that a request the configuration cannot support produces a stated substitution rather than a silent one, on the grounds that a result quietly computed under a different definition than the one requested is indistinguishable from the requested one at the call site.

6 GPU Acceleration

Multiway evolution is a poor fit for the shape of work a GPU is usually given. The frontier is irregular and data-dependent, its width is not known before the step that produces it, and each state must be canonicalized—a refinement loop with data-dependent control flow—before it can be compared with any other. A per-phase kernel launch would put a host round trip between every one of those stages. The design here is therefore a single resident megakernel that keeps the frontier on the device across steps, and the subsections describe what that costs: how canonicalization and matching are expressed without host control flow, how work is queued and termination detected among resident blocks, and how memory is managed when no allocator is available.

6.1 Megakernel Architecture

Traditional GPU algorithms launch separate kernels for each operation, incurring a host round trip between every stage. That cost is fixed per launch and does not shrink with the problem, so it dominates exactly where multiway evolution spends most of its steps: the early frontiers, which are small and numerous. Our *megakernel* approach instead runs a single persistent kernel for the whole evolution, keeping the frontier, the match queues and the epoch counter resident on the device between steps, so a step costs a device-side barrier rather than a launch. Figure 4 shows the arrangement: the queues and the shell that services them, with expanded states re-entering as match tasks without returning to the host. What the design removes, and the depth beyond which that removal no longer registers, are measured in Section 8—it is $16.7\times$ a per-step `evolve()` at depth four and indistinguishable from it by depth eight, once the evolution itself dominates the constant being amortized.

6.2 The Constraint the Design Follows From

The resident kernel exists to remove a host round trip per evolution step. That aim is only met if nothing crosses the host–device boundary during evolution at all: rules and the initial state are uploaded and the queues seeded before the launch, results are read after it, and in between the device decides what work exists, which worker takes it, and when the evolution has finished. A round trip introduced for any other purpose

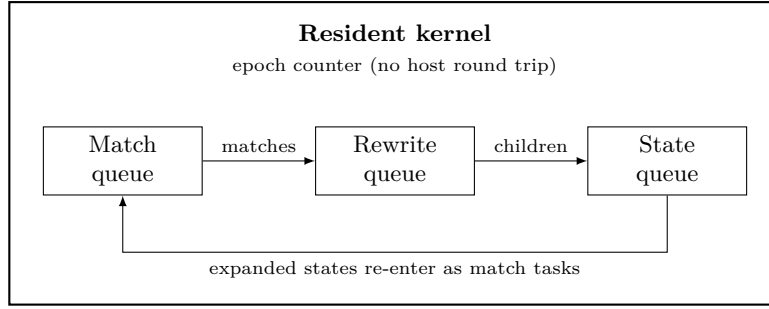


Figure 4: Queue structure of the resident device kernel. Work circulates between the three queues without returning to the host; an epoch counter separates evolution steps.

has the same cost as the per-step round trip the design removes, so the constraint admits no exceptions.

Two properties of the device path follow from it rather than being chosen independently.

Capacity overflow cannot be resolved by retrying with a larger allocation. Enlarging a pool means allocating on the host, which is a round trip. A device-resident loop must therefore record the overflow and return the work completed so far, which is the behavior described in Section 5.7. On the device that behavior is not one option among several; it is the only one available, and it is why capacities are sized from configuration before the launch.

The canonicalizer’s scratch cannot be sized by measuring the workload. Sizing a scratch slot to the largest state in a batch requires knowing the batch, and a resident loop has no batch: states arrive continuously and the largest is not known until the run ends. One resolution is a slot pre-sized from a configured maximum with a coarser hash used above it, which would reintroduce the possibility of two states receiving the same key because one was too large to canonicalize exactly—a silent loss of the exactness property of Section 3, bounded by a configuration value rather than eliminated. The resolution taken instead is a device-side arena from which each state claims a slot sized from its own vertex, edge and occurrence counts. There is consequently no approximate fallback in the device’s exact path: exhausting the arena is a capacity overflow, recorded and reported like any other, and never a quietly coarser identity.

6.3 GPU Canonicalization

The device does not reimplement canonicalization. The individualization–refinement body is a single source compiled for both host and device, so host and device compute the same canonical hash for a given state by construction rather than by two implementations being kept in step. What is device-specific is the orchestration only: how a state is flattened out of the adjacency representation, and where the core’s scratch comes from.

The parallelism is across states, not within one. A worker canonicalizes one state per call, running the shared body serially, which suits a workload whose states arrive continuously and number in the thousands

rather than one whose individual instances are large. The scratch for that call is sized from the state’s own edge and occurrence counts and claimed from the device arena of Section 6.2, so there is no per-state ceiling and no batch to measure.

There is consequently no approximate fallback anywhere on this path. A state the exact path cannot key—because the arena is exhausted, or because the search wants more individualization depth than the device attempts—is reported as a capacity overflow and the run is retried with a larger reservation. It is never keyed by something coarser. This is not a tuning preference, because isomorphism-invariance fails in only one direction: a 1-dimensional Weisfeiler–Leman hash never separates two isomorphic states, but it does merge non-isomorphic ones, which the engine demonstrates constructively on the prism graph against $K_{3,3}$ at six vertices and on the rook’s 4×4 graph against the Shrikhande graph. Nothing bounds how often an evolution reaches such a state, so no collision rate measured over some other corpus would license treating the merge as rare. The only additional device-side hash is the content-ordered key that the AUTOMATIC identity requires, which applies the same rule as the host.

Where a device-side event signature must stand in a raw edge identifier because an edge’s canonical rank was unavailable at the moment the event was stamped, the substitution is counted. Such a signature is not an isomorphism invariant, and the count is what lets a caller comparing event totals across devices see that it occurred rather than infer a disagreement in the evolution.

6.4 GPU Pattern Matching

The unit of matching work on the device is a (state, rule) pair, handled by one block whose threads stripe the candidates for the first pattern edge. Parallelism across states and rules is therefore the outer axis, with the inner axis inside a single join.

The device keeps two indices over the edge pool, a signature index and a vertex inverted index, each filtered by the state’s edge set at query time, and divides the work between them: the signature index seeds the first pattern edge; every subsequent edge is driven by the vertex inverted index through a *pivot variable*—a left-hand-side variable that is guaranteed

to be bound by the time that edge runs, which the connectivity ordering of the pattern edges establishes. Looking the pivot’s binding up in the inverted index yields a candidate list bounded by that vertex’s degree, typically a few entries, in place of a signature bucket that holds thousands on a dense state. This is the division adapted from HGMatch [12]: signature index for the seed, inverted index for the extensions. The host answers the same two questions from the state’s ancestry (Section 4.2).

One detail of the semantics is computed on the host so that the device does not recompute it per match. Because distinct pattern variables may bind to the same vertex, a single pattern signature is compatible not with one data signature but with every coarsening of its partition. The list of compatible signature hashes for each pattern edge is therefore computed once on the host when the rule is added, and the kernel seeds from the index directly rather than enumerating coarsenings while matching.

6.5 Device Work Queues and Termination

The queues of Figure 4 are bounded lock-free multi-producer, multi-consumer ring buffers in device memory. Workers are producers and consumers of the same queues—a rewrite produces states, which become match tasks—so a single-producer structure would not serve, and a bounded one is required because device memory is reserved in advance.

Each slot carries a sequence number, and that number alone determines whose turn the slot is: it equals the position a producer is reserving when the slot is free, one more than that position when the slot holds an item for the consumer reserving it, and less than the position when the queue is full for a producer or empty for a consumer. Slot i starts at i , a producer publishes with the position plus one, and a consumer releases with the position plus the capacity, so a slot advances by exactly one capacity per lap and the three tests remain exact across wraps. The ordering is carried by that handshake—a release store by the producer synchronizing with the consumer’s acquire load—so the head and tail cursors, which only allocate positions, are relaxed.

Reservation is a compare-and-swap on the cursor rather than an unconditional increment. The difference matters precisely because the same workers both produce and consume: an unconditional bump has nothing to undo with, and a rollback or a give-the-slot-back store can land on a slot another thread has already taken, which either loses an item or hands one out twice. In a queue whose producers are its consumers, a lost item is not a dropped unit of work but a run that never finishes, because the termination detector waits for a completion that can no longer occur.

A resident kernel whose workers pull from queues cannot end when a launch ends, so termination is detected rather than assumed. Each queue role maintains two monotonic counters, one for work pushed and one

for work drained, incremented with release ordering so that any payload the work touched is visible to whoever observes the counter. A detector reads them with acquire ordering and looks for every pair to be equal. Equality alone is not sufficient: it occurs mid-evolution whenever every in-flight task happens to have just finished without yet emitting its follow-ups. Exit is therefore signalled only when equality holds across two consecutive observations separated by a backoff interval, so that nothing was pushed during the window between them. Workers test the exit flag at the top of their outer loop, and read it with acquire ordering so a worker that sees it also sees everything the detector saw before setting it.

Detecting termination incorrectly truncates the evolution without an error, so the device result is compared against the host’s on every corpus workload rather than only checked for the absence of reported errors.

6.6 Memory Management

Device memory is reserved before the kernel launches, since a resident kernel cannot call the host allocator. Edge and state pools are pre-allocated with atomic counters, and scratch whose size is known only once a state’s dimensions are seen—the canonicalizer’s working arrays, principally—is carved from a device-owned bump arena.

The arena’s size is determined by the grid rather than by the workload. A persistent kernel’s blocks do not retire and get replaced—they live for the whole evolution—so the grid is the worker count, and each resident worker holds its own arena slot at a time. Arena demand therefore scales with the grid, and sizing it from the state budget alone starves it as soon as the grid grows. The arena is one shared bump pool rather than a partition per worker, because a worker keeps one slot and grows it to the largest state it personally canonicalizes: what has to be provided is the average share across workers, not a per-worker maximum. The default grid is eight blocks per streaming multiprocessor. There is no free operation: a worker finishes with its scratch before taking its next item, so a slot is reused and re-claimed only when a larger one is needed, which with doubling wastes at most one slot per block, bounded by that block’s peak. The whole arena is reset at the end of the run.

Where a pre-sized region proves too small for a state, the overflow is reported through a device-side error channel and the work is retried with a larger reservation, which is the mechanism behind the capacity behavior of Section 5.7.

7 Verification of the Concurrent Surface

The engine holds no lock, so every property that a lock would have enforced is instead a consequence of an ordering argument. Arguments of that kind are checked poorly by testing. A stress test samples inter-

leavings, so a clean run bounds nothing: on this engine a determinism gate of 150 runs – 90 recorded, then 40 and 20 more under load—explores a vanishing fraction of the schedules its own thread count admits, and the schedules that matter are not the likely ones. Two verification layers are therefore maintained alongside the test suite, and they are bounded in different directions.

Bounded programs, all executions. The first layer uses GenMC [28], which enumerates the executions of a bounded concurrent program under the RC11 memory model [29]. For the program it is given—that thread count, that operation count, those inputs—it explores every interleaving and every reads-from choice the model permits, so a clean run is a statement about all executions within the bound rather than about the ones that happened to be scheduled.

Three harnesses carry a weaker guarantee than that, and each says so in place. Two of them bound the number of context switches, which the checker permits only under sequential consistency, so those two establish their property for the sequentially consistent executions of their program rather than for every RC11 execution of it. A third samples executions instead of enumerating them, because its shape cannot be exhausted at the size that exhibits the behaviour it is about; a clean sample is evidence rather than proof, and the same property is proved by enumeration at a smaller shape by a separate harness. The remaining harnesses are exhaustive over RC11 within their stated bound.

Each harness is a small program that includes the engine’s own header and calls its own functions, rather than a hand-written model of the algorithm. The constraint is deliberate: a model drifts from the code as soon as the code changes, and a re-implementation establishes a property of the re-implementation. A harness that includes the header stops compiling when the header changes shape, which couples it to the code it is about. Two harnesses cannot meet the constraint, because the protocols they check live inside functions the interpreter cannot take, and both say so in place: their transcriptions are copied verbatim and run over the real underlying structure.

The properties checked are the ones whose failure would be silent. For the concurrent map: that get-or-create admits exactly one winner and returns that winner to every caller, so a *was_inserted* result means what its callers assume. That this survives a table resize running concurrently with the inserts that triggered it, and survives two successive growths. And that the flag comes from the publishing exchange rather than from a value comparison, so repeated offers of the same value for one key do not each report success. For the deque: that the owner popping the back and a thief popping the front never extract the same item, checked at the single configuration where the two ends address the same slot. For the match rendezvous: that a match is claimed exactly once and never dropped on a collision, and, for the rendezvous between an instance and the matches captured for its class, that

every pair meets. For the per-instance application list: that of any two applications linked into one list exactly one observes the other, permitting the branchial pair they induce to be emitted once with nothing to deduplicate it against (Section 5.10), and that a walk from a node enumerates every node linked before it. For the per-worker scratch arena: that no two workers are handed the same index. For the key set: that two threads inserting one key while a third insertion forces a growth beneath them produce exactly one reported insertion, and that enumeration emits each key once across that growth—two properties checked by separate harnesses, because they are independent and a single harness that bundled them would not say which one a violation belonged to. A third property class covers *readers* racing a growth: a membership query for a key that settled before the query began must answer present in every execution, even mid-carry, when the key’s slot in the retiring table is already sealed and its copy has just landed in the successor. The chain walk is oldest-first for that reason: a newest-first walk visits each table on the wrong side of the carry hand-off and can report absent for a key that is present, and the twenty-execution counterexample distinguishing the two orders is in the harness. The same read-order condition governs the map’s drained-table skip, where the calibrated break is reading a table’s skip flag only when the walk arrives at it. For the scheduler: that a worker which parks is always woken, which is a liveness property and is checked as one—removing either of the two sequentially consistent fences in the submit/park protocol makes the checker report a non-terminating spin loop.

The checker itself. Running an engine of this size through GenMC required changes to the checker, all of which are shipped with the engine as a patch against release 0.17.0. Four are correctness fixes: **operator new[]** is intercepted as an allocation, as **operator new** already was; a **memcpy** is promoted over the element type its destination names, with byte stores for a tail shorter than the last element and alignment-width stores for an opaque **memset**; the read half of a failed compare-exchange carries the failure ordering rather than the success ordering; and no liveness violation is reported while a thread is blocked by an assume, a helped compare-exchange or a confirmation (upstream pull request 58). One is a limitation in the lowering of copies whose operands the promotion pass cannot type—a call result, or a load from a lambda’s closure field—which were lowered as loops at the intrinsic’s alignment, so that a program’s two-byte field read fell inside a four-byte store, an access the checker’s value resolution does not answer and reported as uninitialised memory; the promotion now runs once before and once after scalar replacement and register promotion, when such operands have become the allocas they held, and the report names a read that lies inside a wider write. The rest are instruments: a progress heartbeat over the transformation passes and the exploration, and, in estimation mode, an attribution of the state-space

estimate to the reads and writes that produce it—every access with more than one alternative is booked against its source line with weight $\log_2$ of the alternative count, and the booked total equals $\log_2$ of the estimator’s sample per execution. That attribution is what identified the two changes that made the composed harnesses tractable: an idle worker’s empty pop of its deque wrote its `bottom` word twice per look, giving every thief’s read of that word one reads-from alternative per store, and the workers were spawned with distinct arguments, which kept the checker’s symmetry reduction from applying.

Unbounded worker count, a transcribed model.

The second layer uses TLA+ and its model checker [30] on the match forwarding protocol, which is the property whose silent failure is most costly: a lost match removes the entire subtree below it while leaving the run internally consistent. Every in-flight protocol step is held in a bag from which any enabled step may fire, so the interleavings subsume every worker count rather than a bound on it; what is bounded instead is the state count. The trade is stated where the specification is: this layer is a transcription, so a change to the code does not change the model, and each specification header names the code it transcribes.

The specification is checked in four configurations, differing in whether the claim winner owns both the store and the propagation and whether submission is batched (Table 3). The shipped configuration is forwarding-complete at the model bound. The configuration without the ownership invariant and without batching violates the property. Its counterexample is the shape the invariant exists to exclude. On a chain $s_0 \rightarrow s_1 \rightarrow s_2$, a match is discovered at the root after s_2 ’s pull has run; s_1 ’s pull wins the claim and stores without propagating; the root’s push then finds the claim taken and skips its recursion. That leaves s_2 quiescent with a match that is valid for it and not stored. That the model reaches the failure it exists to exclude is the calibration of the model. The remaining configuration shows that batching alone masks the broken pull at this bound, which is why the ownership invariant is not treated as redundant with it.

Table 3: **Forwarding completeness under TLA+.** The shipped protocol and three calibration arms of the same specification, exhaustive at the model bound of four states, three matches and three edges, with a mid-chain discovery enabling a grandchild. *Ownership* is the claim winner owning both the store and the propagation; *batched* is per-state submission after matching completes. The arm without ownership or batching is the checker’s calibration: the specification detects the violation its removal introduces.

Arm	Ownership	Batched	Verdict	Distinct states
shipped	yes	yes	complete	117,005
calibration	no	yes	complete	85,777
calibration	yes	no	complete	479,005
calibration	no	no	violated (as required)	trace depth 13

A second structure in the same layer. The segmented array carries the states, the events, the edges and the per-instance application lists, and its ordering invariant is that if a segment exists then so does every segment below it—a reader that finds segment k may index any position beneath it without further checking. That invariant is specified rather than tested, because the structure allocates segments on demand from concurrent appenders and the failure it guards against is a read of an unallocated segment rather than a wrong value. The shipped protocol checks clean at 2,284 distinct states and a complete-graph depth of 21; the configuration that creates only the segment asked for violates *DenseBelowCount*. The calibration also fixes the bound: at two appenders the broken protocol is *also* clean, since one claim each means the index that would expose the gap cannot be reached until a holder of the lower segment has published, so three are required. It is specified here rather than checked with the tools of the previous layer because that checker does not survive constructing this structure—a limitation of the tool, recorded where it applies rather than presented as a choice.

What a sampled execution reports. The two layers above bound the schedule; the test suite does not, and no sanitizer changes that. A sanitizer changes what a given execution is able to report, which is the second half of the objection to testing: a stress run samples few schedules, and the schedules it does sample can still pass while holding a race, an out-of-bounds access, or a read of memory nothing wrote. The suite is therefore run under ThreadSanitizer, AddressSanitizer and UndefinedBehaviorSanitizer, and the device suites under the CUDA memory and initialization checkers; all five report nothing on the shipped tree, and all five fail the build on a finding. The initialization checker detects a case the others do not: a host readback of device memory that no kernel wrote returns whatever the allocator last left at that address, so an assertion over it succeeds or fails according to allocation history rather than according to the code, and neither a test nor a model checker over the concurrent structures observes it.

8 Experimental Evaluation

The evaluation answers five questions, and each table below is labelled with the one it addresses.

Q1 (correctness). Is the state identity exact—does the engine identify two states if and only if they are isomorphic—on every workload in the corpus, checked against an independent oracle? (Table 4)

Q2 (cost against the reference). How does evolution time compare with the Wolfram implementation computing the same object, and how does that ratio behave with depth? (Table 6, Figure 5)

Q3 (parallel scaling). How far does evolution scale with worker count, does that limit depend on the

rule, and what resource sets it? (Tables 11, 12, 14; Figure 7)

Q4 (device). Where is the GPU faster than the CPU, and at what grid size does the device saturate? (Tables 8, 9, 15)

Q5 (attribution). Which contributions account for the performance, and does any of them change the computed answer rather than its speed? (Tables 17, 7, 19)

Every table below is written by `tools/dev/paper_tables.py` from the instrument named in its caption, and records the commit and machine it was produced on. The one exception is Table 16, transcribed from an Nsight Compute session because its reports are not text; its caption says so. Captions carry a **T_n** identifier alongside the float number: **T_n** names the generated fragment (`paper/tables/tn_*.tex`) and so ties a table to the tool that wrote it, while the float number is LaTeX’s. The two do not correspond, and the **T** sequence has gaps where a fragment was retired. Numbers are medians over repeated runs in one session; a comparison between two arms is always taken in the same session, because this machine’s run-to-run drift exceeds most of the effects being reported.

8.1 Experimental Setup

Hardware: AMD EPYC 9174F (16 cores, 32 threads, one socket), 503 GB RAM, NVIDIA RTX 4090 (24 GB, compute capability 8.9), on bare metal. The cores are homogeneous, so a speedup column divides by a comparable quantity. Workers are placed by the engine’s own topology-derived order—each last-level-cache domain’s cores and their SMT siblings filled before the next domain is opened, read from the cache topology at start-up on either operating system—which is the placement every user of the engine gets; each table’s provenance line names the machine and the load at the moment the measurement started. A table moved to another machine carries that machine with it. The Wolfram comparison of Table 6 is the only measurement not on this host: it needs a licensed kernel, and its provenance line names the machine it was taken on. The remaining tables and figures were not all produced by one invocation and their provenance lines name several commits, but that spread is bookkeeping rather than a difference in what was measured: no file under the engine’s own source directories differs between any of those commits and the one this document was built from. Table 6 and Figure 5 are the exception in this sense too—they were taken at an earlier commit, across which 114 engine files have since changed—so they are the one measurement here that describes an older engine than the rest.

Software: Ubuntu 24.04, Linux 6.8, GCC with C++20, CUDA 12.0, Wolfram Language 14.3.

Workloads: The rule-type corpus of 17 workloads from `reference/oracle_corpus.hpp` (single/-multi rule; productive, idempotent, reductive, and mixed-arity rules; several initial-state shapes), plus larger depths of the canonical Wolfram rule $\{\{x, y, z\}\} \rightarrow \{\{x, y, w\}, \{y, z, w\}\}$ for the scaling and GPU curves.

Methodology: Wall-clock figures are medians over repeated runs on an otherwise idle machine, with the workers placed by the engine’s topology order over the host’s homogeneous cores; a run taken while the machine was loaded is marked in the table’s provenance line rather than published as though it were quiet. Counts, instruction-level measurements and `getrusage` figures are deterministic and are reported from a single run. One exception is stated where it applies: the left-hand-side sweep of Figure 8 reports the *minimum* of its repeats rather than the median, because contention can only make a run slower, so the minimum is the estimate interference cannot inflate—and that sweep admits a timed run only after checking the machine is quiet.

8.2 Correctness and Memory (Tables T1 and T3)

Correctness comes first because a performance number for a wrong answer is not a result. Every workload in the corpus is evolved in FULL (exact) mode, single-threaded, to its own depth—the `oracle_steps` recorded with it in `reference/oracle_corpus.hpp`, chosen so the brute-force cross-check stays tractable, so counts are comparable down a column and not across rows—and its canonical state set, event set, causal relation and branchial relation are compared against the independent reference implementation. The *Exactness* column must read EXACT for every row, checked against the brute-force isomorphism oracle; all seventeen agree exactly. Table 5 then reports what those runs cost in memory, with arena bytes taken from `bytes_allocated()` rather than RSS because the arena figure is deterministic and reproducible, and what they cost the global allocator, which is the claim the arena exists to support: `cost_matrix` counts every call to global `operator new` during the measured evolution, and allocations per event fall by nearly five orders of magnitude, from 456 to 0.007, from the smallest runs to the largest. The count is fixed set-up performed once per run, divided by an event count that grows while it does not; per-event allocation is served from the arena rather than the allocator. No before/after comparison is given for the contributions that produced this figure: each replaced path was deleted in the commit that replaced it, the work spans dozens of commits, and the instrument that measures arena bytes did not exist at the start of that sequence, so two numbers from two different instruments would not constitute a comparison.

Table 4: **T1**. Per corpus workload: canonical states, events, causal edges, branchial edges, and the oracle’s exactness verdict. Source: `tools/cost_matrix.cpp`.

Workload	Class	Canon. states	Events	Causal	Branchial	Exactness
binary-growth	productive/arity2	36	873	872	0	exact
wolfram-2to4	productive/arity2	45	126	184	111	exact
triangle	productive/mixed-conn	36	124	192	249	exact
reductive-2to1	reductive/arity2	4	15	12	5	exact
idempotent-2to2	idempotent/arity2	1	6	10	0	exact
self-loop	self-loop/arity2	2	1	0	0	exact
mixed-arity	mixed-arity	7	6	5	0	exact
arity3-growth	arity3	65	153	152	0	exact
disconnected-lhs	disconnected	1	126	248	63	exact
arity4-growth	arity4	23	33	32	0	exact
mixed-arity-lhs	mixed-arity/join	2	1	0	0	exact
mixed-arity-init	mixed-arity/init	4	4	0	0	exact
arity3-to-2	arity-reductive	4	4	0	0	exact
arity1-with-binary	arity1/mixed	2	1	0	0	exact
cycle4-automorphic	automorphism	7	68184	109992	68184	exact
star4-automorphic	automorphism	201	68184	56360	0	exact
multi-rule	multi-rule	36	237	214	108	exact

Table 5: **T3**. Arena bytes, heap bytes, and global `operator new` calls during evolution, per corpus workload, ordered by work performed. Source: `tools/cost_matrix.cpp`.

Workload	Canon. states	Events	Arena B	Heap B	Heap allocs	Allocs per event
self-loop	2	1	374904	388136	455	455.0000
mixed-arity-lhs	2	1	391768	388112	456	456.0000
arity1-with-binary	2	1	391792	388096	456	456.0000
mixed-arity-init	4	4	409552	388144	457	114.2500
arity3-to-2	4	4	409668	388176	456	114.0000
idempotent-2to2	1	6	495032	388160	456	76.0000
mixed-arity	7	6	427360	388152	455	75.8333
reductive-2to1	4	15	526160	388208	457	30.4667
arity4-growth	23	33	493952	388056	455	13.7879
triangle	36	124	910192	389680	458	3.6935
wolfram-2to4	45	126	973136	389680	458	3.6349
disconnected-lhs	1	126	1067224	388454	458	3.6349
arity3-growth	65	153	886352	389736	457	2.9869
multi-rule	36	237	1049496	426736	459	1.9367
binary-growth	36	873	2193128	401400	461	0.5281
cycle4-automorphic	7	68184	175961516	1495536	475	0.0070
star4-automorphic	201	68184	106107988	1511904	475	0.0070

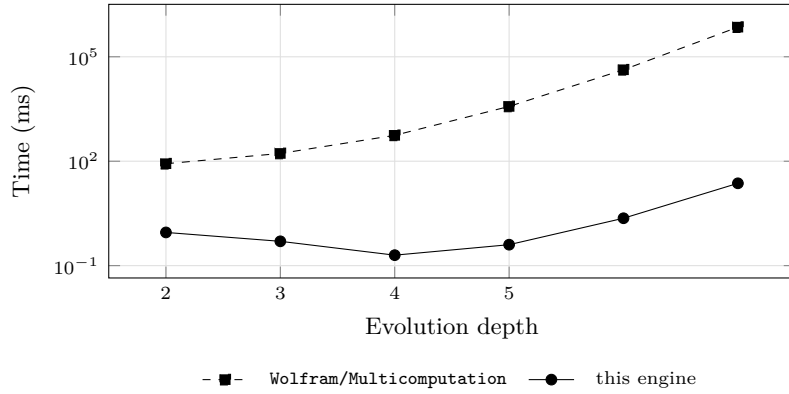


Figure 5: Evolution time against depth for the two implementations, logarithmic in time, on the same rule and initial condition and producing identical state and causal-edge counts at every depth. The vertical separation is the speedup reported in Table 6: $2,770\times$ at depth 4, rising to $30,627\times$ at depth 7. The depth-2 point includes one-time initialization on the Wolfram side, and this engine’s times at depths 3–5 are near the timer’s resolution.

8.3 Wall-Time Speedup (Table T2)

The comparison is against the Wolfram/Multicomputation MultiwaySystem, the published implementation this engine reproduces, and against MultiwayReference, the brute-force Wolfram-language evolver in `reference/`. All three compute the same state and event counts at every depth, without which the ratio would not be a comparison. Over depths four to seven the engine is between $2,770\times$ and $30,627\times$ faster than MultiwaySystem, and the ratio rises monotonically across those four depths. The range starts at depth four because the two shallower rows are not measurements of the same thing: depth two carries the Wolfram side’s one-time initialization, and at depths three to five the engine’s own time is 0.2 to 0.5 ms, close enough to the timer’s resolution that those ratios are lower bounds on the gap rather than precise factors. At depth seven neither caveat applies: the engine takes 23.2 ms against 710 s, so that row is a direct measurement on both sides.

The three are timed at their respective interfaces: the native engine at its C++ API, the Wolfram/Multicomputation paclet’s MultiwaySystem through its exposed properties, and MultiwayReference (`reference/MultiwayReference.wl`) at its function call. MultiwaySystem is the implementation a user would otherwise run; MultiwayReference is a direct, unoptimised evolver of the same definition, and because it returns data rather than Graph objects its time carries no graph construction. MultiwaySystem’s time necessarily does: every property it exposes is graph-valued (`StatesGraph`, `CausalGraph`, `EvolutionGraph`; enumerated by `reference/authority_properties.wls`), and no interface returning the state set alone is available. The table’s *Counts agree* column requires all three to have produced the same state and causal-edge counts at that depth, since a timing comparison between different state spaces is not a comparison; they agree exactly at every depth. The *Engine ms (paclet)* column reports this project’s own Wolfram-layer cost and is excluded from every speedup quoted here: at depth five it is $166\times$ the engine’s own evolution time (66.3 ms against 0.4 ms), and it is the reason no speed claim in this paper is taken through that layer. The speed reported is the engine’s native speed with serialization excluded, the only figure comparable across the three implementations.

8.4 Quotient Exploration (Table T6)

Quotient exploration expands each canonical state once, at its shortest reachable depth, so the number of events actually produced grows far more slowly than the raw count; full expansion and quotient exploration produce the same canonical state set, and the columns report the difference in events produced and in wall time. At depth seven the quotient arm produces 28 events where full expansion reaches 5,913, and reconstruction recovers all 5,913, with their causal edges and branchial pairs, exactly from those 28—the *Exact* column is a per-

row verification, not a description. Reconstruction’s own cost is below the resolution of this measurement: the `quot.+recon` column lands at or under the `quot.` column at most depths, which is run-to-run variation on sub-millisecond timings and not a negative cost. The table establishes that reconstruction is exact and cheap enough not to register beside the evolution it reconstructs from; pricing it would need a workload large enough to separate the two. The saving is in what is expanded, not in what is reported.

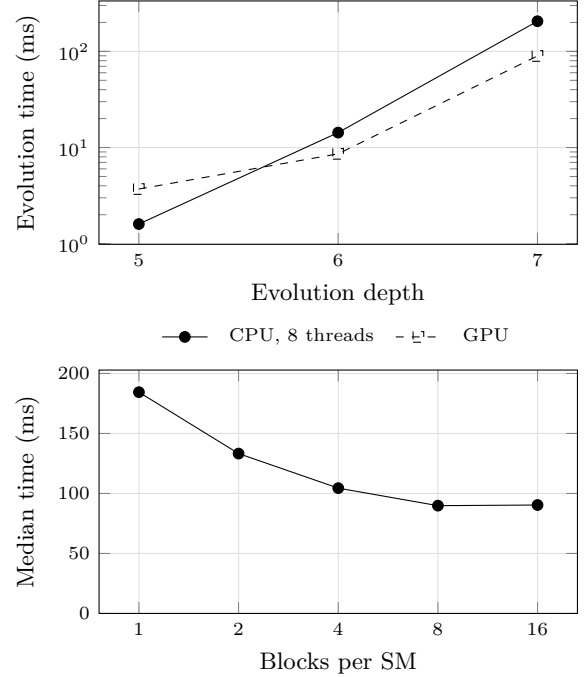


Figure 6: Left: evolution time against depth for the GPU and for eight CPU threads. The device is slower at depth five, where the work per launch does not amortize the launch cost, and faster from depth six. Right: median time against the resident kernel’s grid size. Time ceases to improve beyond eight blocks per streaming multiprocessor, which is the shipped default. Both panels plot the values of Tables 8 and 15, which carry the provenance of each measurement.

8.5 GPU (Table T7)

The device is not uniformly faster, and the range over which it is faster has both ends (Figure 6). Both sides run the same rule, initial condition, canonicalization mode and exploration strategy, and report medians from a single session on the machine in the table’s provenance line; branchial edges are constructed through a per-(state, edge) index. At depth five the CPU wins, because the work per launch does not amortize the cost of getting it there; from depth six the device is ahead of both CPU columns and the margin widens with depth, reaching $2.3\times$ eight CPU threads at depth 7; against a single thread the factor is larger still (Table 8).

The resident-kernel design has its own boundary, on the same workload and measured by the same tool (Table 10). Against a conventional per-step `evolve()` the

Table 6: **T2.** Wall time by depth against `MultiwaySystem` and `MultiwayReference` on `wolfram-2to4`: $\{\{1, 2\}, \{2, 3\}\} \rightarrow \{\{1, 2\}, \{2, 4\}, \{4, 3\}\}$ from $\{\{1, 2\}, \{2, 3\}\}$. Source: `tools/quotient_reconstruction_cost_probe` and `reference/bench_authority.wls`.

Depth	States	Authority ms	Reference ms	Engine ms (paclet)	Engine ms (C++ core)	Speedup vs authority	Speedup vs reference	Counts agree
2	3	84.8	10.6	51.1	0.9	94×	12×	yes
3	4	167.2	36.7	32.6	0.5	334×	73×	yes
4	5	554.0	153.5	42.7	0.2	2770×	768×	yes
5	6	3750.7	835.3	66.3	0.4	9377×	2088×	yes
6	7	42399.9	5589.2	467.8	2.3	18435×	2430×	yes
7	8	710537.0	46944.8	6882.4	23.2	30627×	2023×	yes

Table 7: **T6.** Events and wall time, quotient exploration against full expansion, by depth on the Wolfram rule; *Exact* verifies the reconstruction reproduced full capture’s counts at that row. Source: `tools/quotient_reconstruction_cost_probe`.

Depth	Events (full)	Causal (full)	ms (full)	Events (quot.)	ms (quot.)	Events (quot.+recon)	ms (quot.+recon)	Exact
2	3	4	0.6	3	0.2	3	0.1	exact
3	9	16	0.2	6	0.2	9	0.2	exact
4	33	64	0.1	10	0.2	33	0.2	exact
5	153	304	0.3	15	0.3	153	0.2	exact
6	873	1744	1.6	21	0.3	873	0.2	exact
7	5913	11824	9.9	28	0.7	5913	0.6	exact

persistent evolver is 16.7× faster at depth four, 13.6× at five, 6.4× at six and 4.7× at seven—and 0.96× at depth eight and 0.98× at nine, where the two are indistinguishable. The advantage is the elimination of a fixed per-call cost, which is visible directly in the `evolve()` column: it is 49–55 ms at depths four, five and six, almost independent of a state count that grows from 53 to 3,867 over that range. Once the evolution itself is large enough to dominate that constant—by depth eight, at 262,144 states—keeping the kernel resident removes nothing further, and the two designs are indistinguishable: at depths eight and nine the ratio reads 0.96× and 0.98×, inside the run-to-run variation of Section 8. Reporting the speedup without the depth it is measured at would therefore be a statement about small problems.

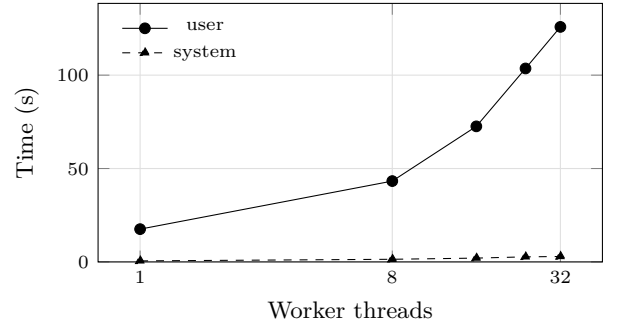


Figure 7: User and system time against worker count, for the runs of Table 14. Across a 32× increase in workers user time rises by 7.2× while system time rises by 5×. System time is thirty to forty-five times the smaller and tracks the fault count. The efficiency decline of Figure 8 is spent in user time, on cross-domain data flow (Table 14’s section), with memory acquisition a minor kernel-side term.

8.6 Thread Scaling (Table T8)

Strong scaling is reported on the Wolfram rule at depth seven, the deepest run Table 8 measures, so that the parallel behavior is read on a run with enough work to parallelize; Table 12 gives the spread across rule shapes. Workers take the engine’s own placement, domain-packed from the cache topology, and the sweep runs at every power of two from one worker to thirty-two, the host’s sixteen cores and their SMT siblings. The table reports speedup over one worker: wall time falls at every step of the sweep while the increment per added worker shrinks, reaching 3.6× at 32 workers. Table 14 identifies what the shortfall at high thread counts is spent on. The expanded state set and the (input, output, rule) transition multiset are identical at every thread count and seed.

8.7 Rule Shape and Where Efficiency Falls (Table T12)

The rule set decides how much parallel work exists, so a scaling curve characterizes the rule as much as the engine. Two shapes are measured at every thread count: a one-edge left-hand side, which re-matches by scanning an index and expands each state narrowly, and a two-edge left-hand side, which joins and produces a wider frontier. With both shapes sized to seconds of single-threaded work, their efficiency curves lie within a few points of each other at every thread count—the floor across the two is 0.37 at eight threads and 0.21 at sixteen—and both keep improving in wall time through thirty-two threads. The difference in how they match, an index scan against a join, does not separate their scaling once each carries comparable work; what governs both is the cross-domain term of Section 8’s memory analysis, which the every-thread-count sweep

Table 8: **T7**. End-to-end evolution time by depth on the Wolfram rule: GPU against CPU at one and at eight threads. Source: `tools/bench_gpu_evolve` and `tools/bench_cpu_evolve`.

Steps	Raw states	Events	CPU 1t ms	CPU 8t ms	GPU ms	vs. CPU 1t	vs. CPU 8t
5	401	400	3.0	1.6	3.7	0.82	0.44
6	3867	3866	34.0	14.3	8.6	3.97	1.67
7	45317	45316	548.2	205.9	89.6	6.12	2.30

Table 9: **T7b** — **GPU against CPU across the named corpus**. The same head-to-head as Table 8, one row per named workload at its standard depth: the single rules, the rule combinations, and the large-state symmetry rows. Both engines run the same rules, initial conditions, canonicalization mode and exploration strategy; medians share one session and one iteration count. Source: `tools/bench_gpu_evolve` and `tools/bench_cpu_evolve` over `corpus_gen.hpp`’s named table.

Workload	Depth	Raw states	CPU 1t ms	CPU 16t ms	GPU ms	vs. CPU 1t	vs. CPU 16t
wpp	7	45317	550.2	182.7	90.1	6.11	2.03
binary	6	22	0.2	0.4	4.1	0.04	0.10
wolfram24	6	9820	104.8	32.3	26.4	3.98	1.23
triangle	6	4	0.1	0.1	2.1	0.03	0.04
arity3	6	7	0.1	0.1	3.6	0.02	0.03
multirule	6	49	20.2	28.1	110.9	0.18	0.25
cycle4	6	40	9.2	11.7	75.8	0.12	0.15
multiroot	6	104	8.7	3.3	13.5	0.65	0.25
disc2x2	6	1409	141.9	64.1	166.4	0.85	0.39
growshrink3	5	8215	52.5	18.3	15.7	3.34	1.16
wolftri	6	10037	112.1	33.9	27.9	4.01	1.21
arimix	6	265	1.6	1.3	5.0	0.32	0.26
allfour	5	25571	230.2	67.9	37.7	6.11	1.80
bigpath	2	512	167.0	23.1	110.2	1.52	0.21
bigcycle	2	514	212.3	27.2	128.1	1.66	0.21
bigstar	2	34	1.3	1.0	11.9	0.11	0.08

Table 10: **T16**. The resident kernel against a per-step `evolve()`, both on the device, by depth on the same rule and initial condition. Source: `tools/dev/persistent_depth_table.py` over `tools/bench_gpu_evolve`.

Depth	States	<code>evolve()</code> ms	Persistent ms	Speedup
4	53	48.9	2.9	16.66×
5	401	49.7	3.6	13.64×
6	3,867	54.6	8.6	6.37×
7	45,317	451.2	96.9	4.66×
8	262,144	927.4	964.1	0.96×
9	524,288	1646.6	1688.0	0.98×

locates at the first step that opens a second L3 domain. The decline is not thread setup: both serial points are seconds, not milliseconds. The table gives the wall times behind the size axis of Figure 8; Section 8.8 sweeps the same axis at four sizes and two further properties of the left-hand side.

8.8 The Three Properties of a Left-Hand Side

Table 12 varies one property of the rule—how many edges its left-hand side has—and holds two others fixed without naming them. A left-hand side has three independent properties, and each changes the matching problem in a different way:

- **Size**: the number of edges to bind. Each additional edge is another factor in the search, and

another constraint that prunes it.

- **Arity**: the number of vertices per edge. Arity changes how much of the state a single edge determines once bound, without changing how many edges must be found.
- **Connectivity**: how the edges share vertices. A path shares one vertex between consecutive edges; a star shares one vertex among all of them; a ring has no endpoint to anchor at; and a left-hand side whose components share *no* vertex cannot be joined at all, so it is matched by enumerating a cartesian product—quadratic in the state size for two components, and worse for more.

Figure 8 sweeps each axis separately, and the result on the first axis runs opposite to the intuition that a larger pattern is harder to parallelize. Parallel efficiency at 32 threads *rises* monotonically with left-hand side size: 0.15, 0.49, 0.67, 0.73 for one, two, three and four edges. The one-edge rule is the worst-scaling shape measured, and it is not scaling badly for want of work: across the four chain sizes of this figure its serial time is the largest. It runs 17.9s, against 5.7, 10.3 and 12.7s for two, three and four edges. (Table 12 sweeps two different rules at different depths, so its serial column is not comparable with these.) A one-edge pattern binds one edge per match, so the work between two synchronizations is small and the fixed cost of publishing a state dominates; each further edge adds join work per match without adding a synchronization.

Table 11: **T8**. Strong scaling: median wall time and speedup over one worker by worker count, Wolfram rule at depth seven, workers placed by the engine’s topology order. Source: `tools/dev/scaling_sweep.py` over `tools/bench_cpu_evolve`.

Threads	Steps	Canonical states	Raw states	Median ms	Speedup vs. 1 thread
1	7	28542	45317	550.333	1.00
2	7	28542	45317	300.490	1.83
4	7	28542	45317	214.849	2.56
8	7	28542	45317	203.908	2.70
16	7	28542	45317	178.822	3.08
24	7	28542	45317	162.602	3.38
32	7	28542	45317	154.592	3.56

Table 12: **T12**. Wall time and parallel efficiency by thread count for a one-edge and a two-edge left-hand side. Source: `tools/dev/scaling_sweep.py` over `tools/sampling_cost_smoke`.

Threads	one-edge LHS ms	one-edge LHS eff.	two-edge LHS ms	two-edge LHS eff.
1	18052.2	1.00	12354.1	1.00
2	9762.6	0.92	6762.2	0.91
4	7056.2	0.64	4824.2	0.64
8	5632.6	0.40	4148.4	0.37
16	4584.8	0.25	3668.8	0.21
24	4356.8	0.17	3503.7	0.15
32	4106.4	0.14	3259.9	0.12

Connectivity moves the curve much further than size does, and in both directions. At 32 threads the ring reaches 1.06 and the branching tree 0.96—superlinear, because a wider frontier lets workers deduplicate states that a serial order would expand—while the path and the hub sit at 0.67 and 0.62. The disconnected shape reaches 0.06: seventeen times worse than the ring on the same machine and the same thread count. That is the widest spread in the whole sweep, and it is the reason connectivity rather than size is the property to state when a rule set’s scaling is being predicted. The disconnected case is reported here rather than in a section of its own, because it is this same axis taken to its limit. Its cost is also not only the cartesian product: on a processor whose cores do not share one last-level cache it is dominated by coherence traffic. Measured on the evaluation machine at a fixed two threads, so scheduling and work distribution are held constant, the same disconnected workload takes 976 ms when both threads share an L3 instance and 1494 ms when they do not, against 1322 ms on one thread—two threads that share a cache speed it up, two that do not are slower than serial.

Edge arity is the weakest axis in one direction and not in the other. At a fixed three-edge left-hand side, efficiency at 32 threads is 0.67 at arity two, 1.015 at arity three and 0.777 for a left-hand side alternating arities two and three. Raising the arity of a path raises the overlap between consecutive edges from one vertex to two, which pins a match’s orientation and removes the second way of binding it; the resulting frontier is narrower per state and deduplicates more, which is where the superlinear value comes from.

8.9 How Far Each Shape Evolves, and How Large Its Relations Become

Each shape is evolved one depth at a time until a single run exceeds the collection budget, so the right-hand end of each curve in Figure 9 is a measured boundary for that shape rather than a depth chosen in advance for all of them. Runs that reached a container ceiling are excluded: the engine returns a truncated graph with a warning when a container fills, and such a run reports where it filled rather than how large the state space is—measured, one shape totals the same 2,097,149 states whether asked for depth 9, 10 or 11, redistributing that ceiling over however many levels were requested. Of the 192 collected points, 41 are excluded on that basis. A shape’s position on the right panel is a property of its rule rather than of how long it was run: the shapes that generate many overlapping sibling matches sit high whatever depth they reached.

Wall time is one property of a workload and the sizes of the objects it constructs are another. Two runs can agree on state count and still differ by more than an order of magnitude in the number of branchial edges they build. At around 300k states the two-component shape reaches 3.2M of them while a mixed-arity chain of comparable size builds *none*, and across the sweep the relation ranges from empty to 58.8M—it is over *pairs* of events sharing a parent, so its size is governed by how many matches overlap rather than by how many states exist. Figure 9 plots both, and Table 13 gives the deepest point reached by each shape: the greatest depth that both fitted the collection budget and did not reach a container ceiling, so the depths differ between rows by construction and are reported rather than held equal. Rows of that table that agree exactly are not duplicates. Raising a path’s arity past three changes nothing the matcher sees, because consecutive edges already overlap in enough vertices to pin a match’s orientation, so `chain2a3` and `chain2a4` describe the same search and report the same counts.

The one-edge left-hand side is the boundary case: it produces *no* branchial edges at any depth, because a single-edge pattern cannot have two overlapping matches at one state to be branchially related. Every shape with two or more edges does.

Table 13: **T14.** The deepest point reached by each left-hand-side shape: **Shape** the connectivity, **Arity** the vertices per edge, **LHS** the pattern’s edge count; the remaining columns are the sizes of the three objects the run built. Source: `tools/dev/rich_sweep.sh` over `tools/sampling_cost_smoke`.

Rule	Shape	Arity	LHS	Depth	States	Causal	Branchial
chain1a2	chain	2	1	7	18,636,561	12,825,936	0
growth	chain	2	1	10	4,037,914	4,037,912	0
chain2a2	chain	2	2	7	9,374,968	15,998,404	12,470,527
pair	chain	2	2	9	21,977,356	42,951,548	19,958,399
chain2a3	chain	3	2	8	6,725,601	11,607,482	5,764,800
chain2a4	chain	4	2	8	6,725,601	11,607,482	5,764,800
chain3a2	chain	2	3	7	4,357,969	12,187,280	12,287,285
triple	chain	2	3	9	21,977,356	64,928,900	37,897,842
chain3a3	chain	3	3	9	12,093,235	34,835,524	18,139,851
chain4a2	chain	2	4	7	1,831,446	7,054,476	7,264,602
quad	chain	2	4	9	21,977,356	86,906,252	53,818,329
cycle3a2	cycle	2	3	14	7,174,453	21,523,347	7,174,452
cycle4a2	cycle	2	4	11	5,592,405	22,369,600	8,388,606
disc2a2	disc	2	2	4	309,853	547,488	3,238,014
disc	disc	2	2	5	17,244,253	31,997,088	214,918,014
disc3a2	disc	2	3	4	1,045,591	3,136,752	58,835,751
mixed2	mixed	2	2	12	22,369,621	42,613,480	0
mixed3	mixed	2	3	14	7,174,453	21,441,385	4,782,968
mixed4	mixed	2	4	14	7,174,453	28,632,164	4,782,968
star2a2	star	2	2	6	6,541,573	12,579,840	24,840,738
star3a2	star	2	3	5	8,967,193	26,779,248	99,436,884
star4a2	star	2	4	5	8,433,241	33,732,864	96,982,260
tree3a2	tree	2	3	7	5,049,431	13,969,772	14,356,591
tree4a2	tree	2	4	7	11,583,495	44,294,548	87,189,943

8.10 What the Lost Efficiency Is Spent On (Table T13)

A falling efficiency curve does not say which resource is being consumed, and the two candidates imply opposite remedies: contention on shared structures would be fixed by changing those structures, while the cost of memory the run acquires would not. Both are read from `getrusage` on the same runs. Across a 32× increase in worker count, user time increases by a factor of 7.2 while system time increases by 5 and stays below three seconds of a hundred and twenty-eight: the cost sits in user time. It is not spinning, because the instruction count is flat across the sweep (the paragraph that follows), and it is not memory acquisition, because the peak resident set grows by a quarter across the sweep while the minor-fault count grows eleven-fold: the kernel term tracks the faults, which come from each worker drawing its own arena blocks, and it stays small. Wall time keeps falling through thirty-two threads on this workload while user time grows seven-fold, so the added workers are busy, and what they are busy with is the cross-domain traffic the next paragraph measures. The result is specific to the host: a part with larger L3 domains would place the two terms differently.

A second term sits in user time, and hardware counters separate it from both candidates above. On the Wolfram rule at depth seven the instruction count is flat across the sweep (14.5G at one thread, 15.8G at 32) while cycles double, so instructions per cycle falls from 1.44 to 0.69; the growing counter is demand fills served from *another* L3 domain’s cache (AMD `ls_dmnd_fills_from_sys_near_cache`), 47k at one thread to 10.6M at 32, while fills from DRAM and from the local hierarchy stay flat. Two experiments

exclude the concurrent structures as the source: capping the shared table’s probe length from 32 to 4 leaves the fill count unchanged while wall time grows, and an exact thread-private filter that removes 80% of the repeated shared-set insertions moves the set’s share of the fills from 36% to 12% without moving the total or the wall time—the traffic re-attributes to the neighbouring sites. The term is therefore the data flow of the evolution itself, about twelve cross-domain fills per object created: a state produced on one core complex is consumed on others, and on a part with eight separate L3 domains those reads are served across domains. Instrument: `tools/dev/perf_fill_sweep.sh`.

The term is visible without counters at the point where it first appears. Sweeping every worker count from one to thirty-two, with workers pinned in order across a part whose L3 domains hold two cores each, wall time falls from one worker to two and *rises* from two to three on every large workload—`wpp` by 12.7%, `wolfram24` by 20.4%, `wolftri` by 12.8%, `multirule` by 36.1%—before falling again from four onward. The third worker is the first to sit on a second L3 domain, and the traffic it opens costs more than its share of the work. Pinning that third worker to an SMT sibling inside the first domain instead removes the rise (`wpp` 347 to 258ms at three workers, below the two-worker time) and moves the same step to the fourth worker, where the second domain opens: the cost belongs to opening a domain, not to a count. Packing each domain’s two cores and two SMT siblings before opening the next is faster at every count measured, including sixteen—`wpp` 197 to 182ms, `wolfram24` 36.3 to 33.3ms, `multirule` 34.7 to 25.7ms—so on this topology halving the number of L3 domains a run touches is worth more than doubling its physical cores. The

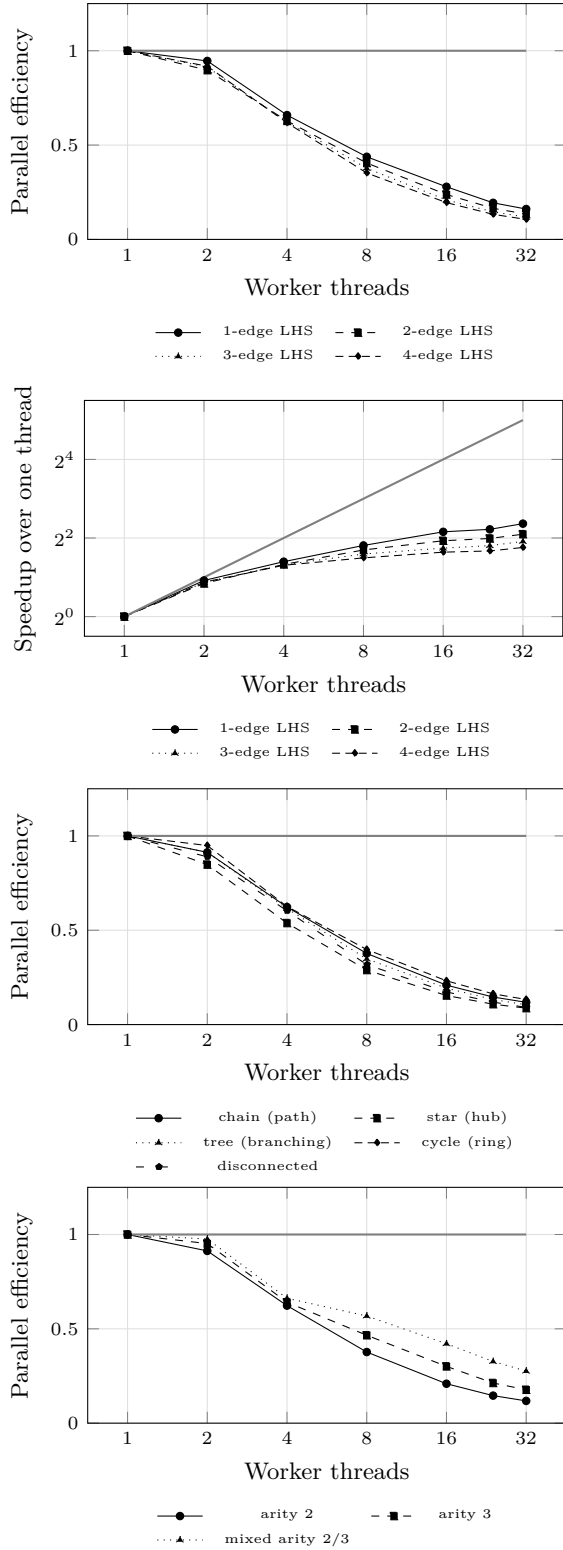


Figure 8: **Parallel efficiency across the three axes of a left-hand side.** Every curve is normalized by its *own* single-thread time, on the same binary, workload and canonicalization mode, so the curves are comparable to each other and none is relative to another system. The grey reference is ideal scaling: $y = 1$ on the efficiency panels, $y = x$ on the speedup panel (upper right, logarithmic on both axes, so ideal scaling is the diagonal). Upper panels: left-hand side size, one to four edges, all binary chains. Lower left: connectivity at comparable size—path, hub, branching tree, ring, and two components sharing no vertex. Lower right: edge arity two and three at a fixed three-edge left-hand side, and one alternating the two. Source: `tools/dev/rich_sweep.sh`.

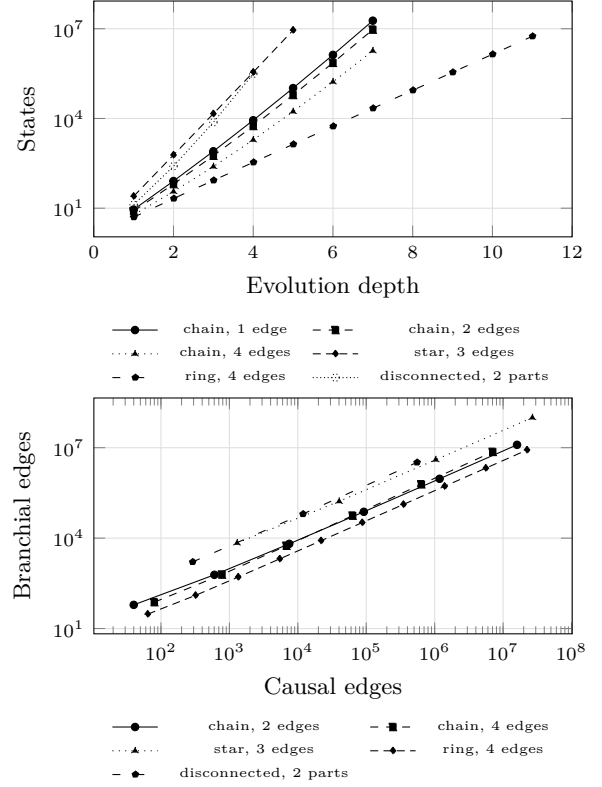


Figure 9: **The state space and the relations over it, by left-hand-side shape.** Left: states against evolution depth, logarithmic in states. Right: branchial edges against causal edges, both logarithmic, every point one (shape, depth) run. Source: `tools/dev/rich_sweep.sh` over `tools/sampling_cost_smoke`.

packed placement is the engine’s own default, derived at start-up from the cache topology the operating system reports, and it is the placement the tables in this section were measured under; the one-per-physical-core order is the counter-example, measured here by naming the CPUs explicitly. Under the engine’s placement the step moves to where the first extra domain opens: on this part a domain holds four hardware threads, so the fifth worker is the first on a second domain and **wpp** runs 216ms at four workers, 247 at five, 234 at six and 205 at eight, after which every further domain opens with the term already amortised (the largest later rise across one to thirty-two workers is 2.3%). No placement of a fifth worker on two-core domains avoids that step, so the supported worker counts on such a part are the powers of two, at which the curve is monotone, and the every-count sweep is the gate that states it. Instrument: `tools/dev/mono_check.py` over `tools/bench_cpu_evolve` at every count, with the placement given as the CPU list.

8.11 Device Saturation (Table T11)

The megakernel is launched at a fixed number of blocks per streaming multiprocessor, so that number is a tuning parameter rather than a property of the problem, and saturation is determined behaviorally: wall time decreases while additional blocks contribute work and flattens when they do not. It flattens at eight blocks per SM, the shipped default, with sixteen per SM within measurement variation of eight; that is where the resident frontier stops being the limit. The table’s final column is the driver’s peak SM-busy reading during the run. It is an upper bound on utilisation and not an occupancy figure, since the counter reports whether the SMs were busy rather than how many warps were resident; its mean is not reported because it decreases as the run becomes faster—the same busy kernel occupies a smaller fraction of the sampling window—and therefore tracks duration rather than residency. That bound reads 100% at every point, so the flattening is not the bound being reached.

8.12 Occupancy, Lane Efficiency, and the Bounds They Set

Table 15 establishes saturation behaviorally and states that its final column is not an occupancy figure. This section supplies the occupancy figure and the second bound that sits beside it, both read from the shipped binary rather than inferred from timings, so that the device results above are reported against the ceilings they were measured under.

Two resources bind the resident kernel, and they multiply rather than compete.

Registers. Reading the linked binary with `cuobjdump -res-usage, k_persistent_evolve` is allocated 255 registers per thread, which is the architectural maximum. The kernel is launched with 32 threads per block, so a block reserves $255 \times 32 = 8,160$ registers; a streaming multiprocessor on the `sm_89` part used

here provides 65,536, admitting eight resident blocks, or 256 threads against a hardware maximum of 1,536. Occupancy is therefore 16.7%, and it is register-bound rather than bound by shared memory (48 bytes per block) or by block count. Per-translation-unit object files report no stack or local frame for any function under separate compilation (`-rdc=true`); only the linked binary carries the allocation, and reading the intermediate objects understates it to zero.

Lane efficiency. The canonicalization stretch of the kernel—the phase the measurements below show to dominate—runs on the whole warp: every lane enters the shared canonicalization core together under an execution policy, the order-safe loops of refinement fan across the lanes, and the search’s serial trunk (individualization, the orbit closure on return, the backjump) executes on the leader behind warp barriers while the other lanes hold the same control flow. The publication, signature, quotient and deduplication steps that follow it run on the leader alone. The achieved width is measured rather than derived: on **wpp** at depth six the profiler reports 3.5 active threads per warp averaged over the kernel, against a ceiling of 32 (Table 16). That average is the two regimes weighted by their cycle share, and the serial trunk is the larger share.

Hardware counters exclude every memory and arithmetic alternative, and a direct experiment excludes occupancy itself. No unit is near its limit: on **wpp** at depth six SM throughput is 9.7% of peak, memory throughput 24.3% and DRAM 11.6%, and across the corpus the maxima are 12.6%, 33.8% and 16.1% (Table 16). Atomic traffic, which the work queues make the natural suspect, lands in L2, whose throughput sits at that memory figure, leaving several times’ headroom.

Occupancy requires more care than the register figure alone suggests, because two independent limits coincide. The resident kernel is allocated 255 registers, which over a 32-thread block permits eight blocks per streaming multiprocessor; the kernel is also launched with a grid of eight blocks per multiprocessor. Both give eight warps of a possible forty-eight, or 16.7%, and the measured occupancy of 16.7% on **wpp** (Table 16) is consistent with either. They are separable by experiment. Capping the whole device library at 128 registers halves the allocation and leaves occupancy unchanged, so the register ceiling was never the binding constraint; raising the grid with that cap in place doubles residency and makes the kernel *slower*, degrading further as the grid grows. This is consistent with the behavioral saturation of Table 15 and identifies its cause: additional resident warps contend for the shared work cursors without contributing independent work.

The device is therefore limited neither by bandwidth, nor by queue contention, nor by occupancy. It is limited by the dependency structure of the work itself. Individualization–refinement is a depth-first backtracking search in which each step depends on its predecessor, and no quantity of resident parallelism executes a dependent chain faster. The scans over vertices and edges are the loops the warp fans; the search itself fans nothing. With the scans distributed, the phase’s remaining

Table 14: **T13.** User and system time, peak resident set and minor page faults against thread count, from `getrusage`. Source: `tools/dev/scaling_sweep.py` over `tools/sampling_cost_smoke` under `/usr/bin/time -v`.

Threads	Wall s	User s	System s	Peak RSS MB	Minor faults
1	18.10	17.55	0.55	5906	70 058
8	5.67	43.28	1.41	6217	254 208
16	4.79	72.58	2.04	6626	457 700
24	4.59	103.58	2.71	6950	614 126
32	4.19	125.83	2.90	7261	769 543

Table 15: **T11.** Wall time and peak SM-busy against blocks per streaming multiprocessor. Source: `tools/dev/scaling_sweep.py` over `tools/bench_gpu_evolve` mode 2.

Blocks per SM	Blocks	Median ms	vs. 1 per SM	Peak SM-busy
1	128	184.40	1.00×	100%
2	256	133.23	1.38×	100%
4	512	104.43	1.77×	100%
8	1024	89.82	2.05×	100%
16	2048	90.36	2.04×	100%

cost is the chain, and the chain is the device’s ceiling for this algorithm on this architecture—a property of canonicalization rather than of the implementation.

Taken as a product, a small fraction of the device’s arithmetic capability is reachable by the dominant phase. The measured throughput reflects that bound, and it is a structural property of the current work decomposition rather than a scheduling artifact or a stall: an independent instrument (below) reports zero time spent waiting on unpublished work.

Per-phase attribution, taken from `clock64` deltas accumulated per block and summed at exit, locates the cost: on the corpus workloads that generate enough work to attribute anything, canonicalization takes the large majority of block cycles, rewriting a few per cent and matching under two. The remaining corpus shapes produce between four and 1,409 states at depth six (Table 9) and are idle most of the time; that is the absence of work rather than a property of the scheduler, and their idle fraction should not be read as a device inefficiency. Canonicalization is thus both the dominant phase and the phase whose serial trunk sets the lane average, and the two bounds above identify the same site in the code by independent routes.

A third bound sits outside the kernel, in the per-call bracket rather than in the evolution itself. Tracing a device run and placing every allocation and copy relative to the resident kernel’s own execution: no copy, allocation or memset occurs while the kernel is in flight—the count is zero across every evolution measured—so the evolution proper is free of host traffic, as the design requires. The bracket around it is not: each evolution’s kernel is preceded and followed by runtime allocation, copy and fill calls, in equal numbers on the two sides of the launch. The design goal of a single readback at the end is therefore not met, though the stronger property of no traffic during evolution is. This bounds interactive and session use, where a caller pays that bracket on every call, rather than throughput on a

single deep run, where it is amortized over the kernel’s duration.

Every device figure above is measured with the record set off, and that configuration hides a second regime. The reconstruction of the raw causal and branchial relations is selectable independently of the evolution, and the measurements reported so far switch all of it off. Re-measured with causal recording on, the device and host agree *exactly* on the reconstructed relation for the eight corpus workloads the device backend supports, at depth six—identical distinct-pair counts and identical counts surviving transitive reduction. The two implementations share no code below the shared replay core, so the agreement is between two separate schedulers, memory layouts and queueing disciplines. On the densest workload available, a three-edge disconnected left side at depth three, the two engines agree on 1,661,760 causal pairs and 970,560 surviving reduction.

Reaching that agreement is a property of how the reduction is defined rather than of how it is computed. A finite directed acyclic graph has exactly one transitive reduction, so membership is a function of the finished relation; both engines therefore store the relation and reduce it on demand, rather than deciding each pair as it arrives against the pairs seen so far. The two engines share that body. Deciding on arrival makes the answer depend on the schedule: a pair reaching the relation before the longer path that bypasses it is kept, the same pair reaching it afterwards is dropped, and an append-only adjacency never retracts the difference. The reduction is computed once per source rather than once per pair—every edge leaving a given event shares the set reachable from it in two or more steps—and where event identifiers increase along every causal edge the search is bounded by the largest target of that source. The branchial relation is handled the same way: a branchial pair is two applications of one instance sharing a consumed slot, so neither engine stores the pairs, and both derive them from the applications on demand.

The idle fraction of the low-work shapes is not, in general, the absence of work. The attribution above reads the idleness of the corpus shapes producing few states as work not existing. With causal recording on this reading no longer holds for two of them: at depth six `cycle4` and `multirule` reconstruct causal relations orders of magnitude larger than their state

counts, and both remain almost entirely idle. Hardware counters place them far from every ceiling—SM throughput 2.2% of peak and DRAM 0.01% on `cycle4` (Table 16)—and their lane efficiency is among the corpus’s best, 9.8 and 10.4 of 32, against 2.7 on `wolfram24`, which the device completes faster than sixteen host threads (Table 9). The decisive comparison is that the kernel takes four times longer on `multirule`’s 49 states than on `wolfram24`’s 9,820.

What separates the two regimes is the parallel width, and the corpus supplies a controlled comparison for it. At depth six `cycle4` replays 68,185 instances over 40 classes and loses to the host; `wolfram24` replays 86,086 over 9,820 classes and beats it (Table 9). The instance counts are within 30% of each other and the outcomes are opposite, so neither the amount of replay work nor the depth of the recursion is the discriminator. The state count is: a class is the unit over which the replay’s instances are independent, and a run that collapses to forty classes has forty-fold width available however much work those classes carry. `cycle4` begins from a deliberately automorphic initial state, which is exactly the condition that collapses classes. The device’s disadvantage on these shapes is therefore a shortage of independent work rather than an inability to execute the work it has, and raising occupancy against it would raise the idle count rather than the throughput.

The instances *within* a class can be replayed concurrently. They are serialized by the worklist: each driver takes its instances from a worklist private to it, so a class is walked by one driver and, on a run that collapses to forty classes, forty such walks proceed while the remainder of the grid spins.

The instance count and the class count are far apart on exactly the shapes that lose. At depth six `cycle4` materialises 68,185 instances over 40 classes and `multirule` 146,599 over 49—deficits of $1,705\times$ and $2,992\times$ —while `wolfram24` materialises 86,086 over 9,820 and `wpp` 15,967 over 3,867, deficits of $8.8\times$ and $4.1\times$. The instance counts are comparable across all four, between 16 and 147 thousand, while the class counts vary by a factor of 250: the independent work available is similar everywhere and only the width the implementation exploits differs. The shapes that lose are those whose initial state collapses classes, and that collapse is the same condition that concentrates instances into few of them.

The residual on these two workloads is accordingly attributed to a named and evidenced cause: the replay’s depth-first descent runs inline on the discovering thread, so a class’s instances are walked by one driver, and a run that collapses to forty classes has forty drivers’ worth of width however many instances those classes carry. That is a bound of the replay’s work decomposition, stated as measured, and it is not claimed as a lower bound on the problem.

Bounds these results were measured against. Nothing in this corpus is near a hardware ceiling: across the profiled corpus the highest SM-throughput figure

is 12.6% of peak (`wolfram24`) and the highest DRAM figure 16.1% (`growshrink3`) (Table 16). Occupancy is structural for the reason given above. The lane average separates the two regimes: about three lanes per warp on the large-state workloads, where the canonicalization trunk binds, and nine to twelve on the shapes of at most forty-nine states, where the shortage of independent work binds; `disc2x2`, `arimix` and `multiroot` sit between, at six to nine. The allocation invariant holds with the reconstruction running as well—profiling twenty-four consecutive evolutions records zero allocations, copies or memsets between a resident kernel’s start and its end, with every such call falling in set-up or read-back. What is not established is whether the serial chain is irreducible; it is attributed and bounded, not proven minimal.

The quotient decoupling of Section 5.10 applies to the device on the same terms as the host. Capture of each class’s frame is separated from replay of that capture against instances. The first signs event identity at the cost of the canonical answer, the second recovers the raw event set at the cost of the raw one. Together they leave the canonical answer bit-identical while removing the exponential from runs that do not request raw events. Two corpus shapes, `multirule` and `cycle4`, do not complete depth seven under unconditional replay at all; with capture and replay separated both complete, with state and event counts identical in the two arms, and the step from depth six to seven costs almost nothing further where the unconditional arm does not terminate.

Two routes around the canonicalization bound were measured and rejected. Both are reported here because they bound the same quantity from two sides, and because the two fail for opposite reasons. Tiered canonicalization—bucket by the Weisfeiler–Leman signature, run individualization–refinement only on a collision—was implemented and measured at 28% slower, because a bucket hit still requires refinement to confirm isomorphism, so a duplicate state is processed by both stages. A content-equality short-circuit reuses a canonical form when two raw states hold literally the same edge tuples. It needs no confirmation, being a proof of isomorphism on its own—and it never fires. Measured over `wpp`, the number of distinct state contents equals the number of raw states exactly at depths six and seven (3,867 of 3,867; 45,317 of 45,317), because every rewrite mints fresh vertex identifiers, so no two states reaching one canonical form carry the same labels. The measurement is order-independent and therefore an upper bound on what any content key could catch. The two routes fail oppositely: the first fires often but confirms redundantly, the second would need no confirmation but never fires. The $1.59\times$ gap between refinement passes performed and distinct canonical states is real; reaching it requires recognising isomorphic states without canonicalizing them, which is the problem canonicalization solves.

These figures hold across the corpus rather than on one shape. Profiling the resident kernel on every corpus workload—measured with the profiler rather than by

wall clock, so that occupancy, throughput and the lane average are read from hardware counters rather than inferred from elapsed time—gives occupancy between 14.1% and 17.3% on every one, over workloads whose profiled kernel durations differ by a factor of seven hundred (Table 16):

Table 16: **Resident-kernel occupancy, throughput and lane average across the corpus**, read from hardware counters under NVIDIA Nsight Compute: means over the profiled launches of `k_persistent_evolve`. Durations are the profiler’s, which serializes execution and fixes clocks, not wall figures. Source: `tools/dev/ncu_occupancy_table.py` over the reports the floor phase captures.

Workload	Depth	Launches	Duration (µs)	SM %	Mem. %	DRAM %	Lanes/warp	Occ. %
allfour	6	6	187381.7	8.46	21.51	9.75	3.00	17.03
disc2x2	6	7	178994.3	3.88	3.95	0.17	6.05	16.84
multirule	6	5	113454.0	2.24	2.30	0.02	10.41	17.25
bigcycle	2	9	86642.2	7.93	9.11	0.15	2.83	16.65
bigpath	2	11	82264.5	6.71	9.14	0.17	3.16	16.66
growshrink3	6	12	81015.0	10.86	33.27	16.08	3.14	16.99
cycle4	6	3	73393.3	2.23	2.72	0.01	9.84	16.39
wolfram24	6	3	17210.0	7.46	20.83	8.46	2.74	14.12
multirule	6	3	13420.0	2.38	2.43	0.04	8.87	15.54
wolfram24	6	3	9086.7	12.59	33.75	12.88	2.71	16.41
bigstar	2	3	8806.7	2.44	2.43	0.01	9.79	16.73
wpp	6	3	3900.0	9.65	24.25	11.57	3.45	16.71
arimix	6	3	3213.3	2.99	4.29	0.21	8.08	16.58
binary	6	3	2676.7	2.20	2.22	0.03	11.17	16.77
arity3	6	3	2106.7	2.20	2.17	0.03	11.59	16.42
triangle	6	3	263.8	2.60	2.95	0.21	11.59	16.28

A grid-determined limit produces exactly this: an occupancy constant across a corpus this varied, and confirms from a third direction that it is not determined by register allocation.

The duration column is the profiler’s and is read as such: the profiler serializes execution and fixes clocks, and it excludes the host-side bracket around the launch, so a profiled duration is neither the wall time of Table 9 nor comparable to it. The ratio columns are unaffected. Those are structural properties rather than elapsed times, and the table reports them for that reason: an occupancy between 14.1% and 17.3% across the corpus, utilisation figures that never approach any unit’s limit, and a lane average that separates the two regimes named above. The shapes with a large state population run near three active lanes per warp—`wolfram24` 2.7, `growshrink3` 3.1, `wpp` 3.5—because their blocks spend their cycles in the canonicalization trunk; the shapes of at most forty-nine states run at ten to twelve lanes, because their blocks spend most cycles in the fanned match phase and in the idle loop, and lose to the host anyway for want of independent work.

We report these as bounds. The canonicalization phase is distributed across the warp to the extent its loops admit, and the register allocation is at the architectural maximum; the occupancy experiment above shows that residency raised on its own slows the kernel, and the profiler’s lane average shows where the remaining cycles sit. The ceilings are stated here as measured, with the mechanism that sets each one named.

8.13 Ablation (Table T9)

Only two of the engine’s contributions remain run-time switches; the rest were replaced outright and their predecessors deleted in the same commit, so no “off” arm

exists to measure. The two are match forwarding and the state-canonicalization mode; quotient exploration is reported separately in Table 7. Each arm is a best-of- N wall time. Match forwarding is decided per rule set by the static analysis of Section 5.6 rather than being switched on globally. On the one-edge left-hand side (depth eight) the analysis decides it off and the forwarding arm is $1.16\times$ slower, so the decision is the measured one; on the two-edge left-hand side (depth six) the analysis decides it on and that arm is $1.10\times$ slower than with forwarding off, a margin of the size this box’s interleaved medians swing on runs of that length (Section 8). State and event counts are compared across arms, so a switch that alters the computed result is identified as such rather than reported as a change in speed. That column is a statement about these workloads alone: the canonicalization mode does change the computed answer on other inputs—Section 5.15 gives a corpus case where the exact and approximate modes report different event counts—and the row reports agreement because the two-edge left-hand side at depth six is not such a case; the workload is named because the agreement claim holds only for it.

The canonicalization row is the one that cuts against the framing of Section 3. Exact individualization—refinement is not the expensive option here: it runs at 39.6 ms against 52.8 ms for the cheaper content hash, a ratio of 0.75. The hash does not remove the refinement pass, it adds a second key on top of it—the sampling draw is keyed on `canonical_transition_key`, whose first act is the same refinement—so on this workload the approximate mode pays for both. That is the same result Proposition 3.3 predicts from the other direction: a cheaper filter in front of IR cannot remove the IR invocations, so it can only add to them.

8.14 Where the Instructions Go (Table T15)

Wall time says how long a run took; it does not say which part of the engine spent it, and on a machine whose clock drifts between sessions it is the wrong instrument for attribution. Executed instruction counts are deterministic, so this table is taken once under `callgrind`, on a single-threaded depth-six evolution of the Wolfram rule under quotient exploration with the raw relations reconstructed—the configuration `bench_cpu_evolve` runs by default—and needs neither repeats nor a quiet machine. The quotient reconstruction (23.5%) and canonicalization (22.9%) are the two largest components and are close; the concurrent sets and maps (13.4%) and libc’s copies, fills and heap traffic (13.2%) follow; matching, its join and the rewrite together are 5.6% on a rule whose states carry at most fourteen edges. The unattributed share (5.4%) is functions below the reporting threshold and code no bucket names, and it is reported rather than distributed over the named buckets.

Against bounds. Each host phase is divided by the unit it produces once per item, read from the same run’s

Table 17: **T9**. Ablation over the two run-time switches in the shipped binary: match forwarding (one-edge LHS at depth eight, two-edge at depth six) and the state-canonicalization mode (two-edge at depth six). Best-of- N wall times; the counts column certifies both arms computed the same result. Source: `tools/sampling_cost_smoke`.

Contribution	Arm A	A ms	Arm B	B ms	Ratio	Same answer
Match forwarding, one-edge LHS	on	70.7	decided off	61.0	1.16	yes
Match forwarding, two-edge LHS	off	35.6	decided on	39.1	0.91	yes
State canonicalization	Full (exact)	39.6	Automatic (hash)	52.8	0.75	yes

Table 18: **T15 — Executed instructions by component**. A single-threaded depth-six evolution of the Wolfram rule under quotient exploration with reconstruction, under `callgrind`, grouped by the algorithm each function belongs to. Instruction counts are exact and reproducible, so attribution uses them rather than wall time. The flat per-function profile covers 99.0% of the program total; the remainder is below `callgrind_annotate`’s reporting threshold. The benchmark-harness row is the corpus set-up the bench performs before the evolution. Source: `tools/dev/instruction_profile.py` over the `attrib` phase of `tools/dev/remote_session.sh`.

Component	Instructions	Share
Quotient reconstruction (replay, causal DP, TR)	66,492,778	23.5%
Canonicalization (IR and orbits)	64,694,936	22.9%
Concurrent sets and maps	37,880,185	13.4%
libc copies, fills and heap	37,468,155	13.2%
Arena allocation	21,222,021	7.5%
Hypergraph storage	17,031,469	6.0%
Match expansion, join and rewrite	15,812,810	5.6%
Benchmark harness	3,830,116	1.4%
Job system and work stealing	453,078	0.2%
Unattributed	15,175,538	5.4%
Total	282,876,709	99.0%

counters: 3,881 canonicalization calls over 3,867 raw states, 15,966 reconstructed events, 182,798 claim-set inserts and 3,866 applied matches. Canonicalization costs 16,670 instructions per call on states of two to fourteen edges. The search opens a second node on 28 of the 3,881 calls, so the floor is one refinement to a stable partition over the state’s incidence structure, a few thousand instructions at this size, and the measured cost is several times that: the profile places 25% of the phase in the refinement loop, 51% in the per-call set-up and invariant hash around it, and 24% in the orbit computation the quotient’s event identity requires. Reconstruction costs 4,165 instructions per reconstructed event against a floor of one identity claim, a producer lookup per consumed edge and a producer record per produced edge, a few hundred instructions for a two-in, four-out rule; the excess sits in the per-instance replay descent (`qr_apply`) and in `register_quotient_transition`’s vector traffic. The claim sets cost 207 instructions per insert against a floor of one hash, one probe and one exchange, around fifty; 39.7% of the inserts offer a key the set already holds and pay the probe for nothing. Matching, the join and the rewrite cost 4,090 instructions per applied match, a figure that includes building the match record and the child state, so its floor is the child’s

construction rather than the two edge bindings. No host phase is within a factor of two of its floor, and each excess is attributed above to a named site. The libc share is copies, fills and heap traffic; the raw profile attributes its largest callers to the benchmark’s own set-up, inlined into `main` under link-time optimisation, and to the construction of rule objects.

8.15 Event Canonicalization Overhead

Event identity is not one convention but three, and they disagree in general rather than in corner cases. Table 19 evolves each workload once, under FULL state canonicalization and single-threaded, and counts the distinct events under each: identifying an event by its endpoint states, by the edges it consumed and produced, and by the distinct rule applications. The conventions refine one another—two rewrites are the same event when they run between the same states, when they additionally move the same edges, or never—so the counts are non-decreasing left to right, and where they strictly increase the finer convention separates rewrites the coarser one merges. On `binary-growth` the three give 5, 8 and 9. The table carries no timing column: what separates the conventions is what they identify, and the cost of the identity is the added canonicalization pass Table 17 measures. A paper reporting an event count must therefore say which convention it used, and this one uses the canonical convention throughout.

8.16 Limitations

Single machine. All numbers were taken on one desktop, whose configuration is given in Section 8 and repeated in each table’s provenance line. The scaling results in particular are properties of this machine’s memory system and core count, and the thread count at which efficiency declines is not portable to a machine with a different one.

Quotient reconstruction scales with the system it avoids visiting. Quotient exploration visits one representative per isomorphism class, and its state count can grow far more slowly than the raw system’s. The reconstruction that recovers raw observables from those representatives does not share that saving: it materialises instances, and there are as many instances as the raw system holds. Measured on a two-rule set whose canonical state count grows linearly with depth, the reconstruction’s cost grew about twentyfold per step—0.17, 0.22, 0.62, 7.36, 144.78 and 4981.05 milliseconds at depths two through seven for three, four, five, six, seven and eight canonical states. The growth

Table 19: **T10.** Distinct events per identity convention on each corpus workload. Source: `tools/mode_matrix_probe.cpp`.

Workload	Steps	Canon. states	By endpoint states	By consumed/produced edges	Distinct applications
binary-growth	3	6	5	8	9
wolfram-2to4	2	4	3	4	6
triangle	2	3	2	4	4
reductive-2to1	3	4	3	6	15
idempotent-2to2	3	1	1	1	3
self-loop	3	2	1	1	1
mixed-arity	3	4	3	3	3
arity3-growth	3	9	9	9	9
disconnected-lhs	3	1	1	4	14
arity4-growth	3	9	9	9	9
mixed-arity-lhs	3	2	1	1	1
mixed-arity-init	3	4	4	4	4
arity3-to-2	3	4	4	4	4
arity1-with-binary	3	2	1	1	1
cycle4-automorphic	3	4	3	15	144
star4-automorphic	2	5	4	12	24
multi-rule	2	10	11	11	11

rate is a property of the output rather than of the implementation: the raw observables are as numerous as the raw system holds, whatever computes them. The cost applies only to runs that request those observables. The reconstruction is enabled by the request, so a run asking for canonical states alone does not perform it.

The reconstruction does not scale with threads. Every worker drives it, and the total cycles it consumes grow with the thread count—measured at 511 million cycles on one thread against 3.85 billion on thirty-two for the same workload, with the same amount of admitted work. The dedup marks prevent duplicate effects but not duplicate traversal, so added threads contend rather than divide the work. On that workload wall time improves to eight threads and degrades beyond it.

Two comparison points, one of them our own. The implementations compared against are the `Wolfram/Multicomputation` packet’s `MultiwaySystem` and this project’s own reference implementation. The first is the only external implementation known to us that computes the deduplicated states graph under an isomorphism identity; `SetReplace` computes a different object (Section 2), so no wall-time comparison against it is reported. The second is written by the same author as the engine, from the same definition, which makes it an independent implementation of the definition but not an independent party’s implementation.

Workload scope. Two corpora are used and they are different sizes. The oracle corpus of `reference/oracle_corpus.hpp` is 17 rule-type workloads and is what Tables 4, 5 and 19 report; the left-hand-side sweep of Section 8.8 generates 24 shapes of its own, reported by Table 13. Both were constructed to cover rule shapes—productive, reductive, idempotent, mixed arity, disconnected and high-automorphism—rather than to sample rules found in practice, and the observed behavior of C/R and of parallel efficiency is a property of those shapes.

Scope of the concurrency results. The model-checked properties hold for the thread and operation

counts given, which are small; they are not proofs for unbounded thread counts. The determinism gate likewise establishes agreement across the configurations it runs, not for all.

Worst-case cost. Individualization–refinement is exponential in the worst case, and the exact mode applies it to every state. States with large automorphism groups are where that cost is realized, and Proposition 3.3 shows that a cheaper pre-filter cannot avoid it.

9 Conclusion and Future Work

We have presented an implementation of multiway hypergraph evolution built on exact canonicalization and lock-free concurrency. Compared with the `Wolfram/Multicomputation` packet’s `MultiwaySystem` on the same rule and initial condition, and producing identical state and causal-edge counts at every depth, the engine is faster by a factor of 2,770 at depth 4, rising to 30,627 at depth 7 (Table 6). The ratio rises monotonically across those four depths. At the shallow end of the range the engine’s own time is 0.2 to 0.5 ms, close enough to the timer’s resolution that those ratios are better read as lower bounds on the gap than as exact factors; at depth seven it is 23.2 ms against 710 s, so that row is a direct measurement on both sides. The same table compares the engine against `MultiwayReference`, this project’s own brute-force evolver, which computes the same object from the same definition and returns data rather than graph objects, so that ratio is not attributable to graph construction on the slower side; all three agree on state and causal-edge counts at every depth. On the host, evolution attains a speedup of $3.6\times$ at 32 worker threads, with wall time falling at every step of the sweep (Table 11). Efficiency remains at or above 0.37 up to eight workers and at or above 0.21 up to sixteen, depending on the rule’s shape, and declines beyond that (Table 12). The measured cause is cross-domain data flow, not contention and not memory acquisition: across

a $32\times$ increase in worker count the instruction count stays flat while instructions per cycle falls from 1.44 to 0.69, the growing hardware counter is filled served from another L3 domain’s cache—the evolution’s own producer-to-consumer traffic, with the concurrent structures excluded by direct experiment—and user time grows by $7.2\times$ against a kernel-side term (system time by $5\times$, tracking the fault count) that stays below three per cent of the total (Table 14). On the device, wall time ceases to improve beyond 8 blocks per streaming multiprocessor (Table 15), and the GPU is $2.3\times$ faster than eight CPU threads at depth 7 (Table 8). The contributions are:

1. Exact canonicalization via individualization–refinement (IR) computed on every state, its hash serving directly as the deduplication key, validated against a brute-force isomorphism oracle
2. Lock-free concurrent data structures, whose properties are model-checked rather than stress-tested (Section 7)
3. A resident CUDA kernel that carries matching, rewriting and canonicalization without returning to the host between evolution steps
4. Independent, exact state and event equivalencing axes with configurable granularity
5. Quotient exploration, transition-level sampling, and an online transitive reduction of the causal graph, each preserving the observable contract of Section 5.8
6. A determinism contract: the state set, event set and causal and branchial relations are a function of the inputs alone, independent of thread count and scheduling order (Section 5.8)

9.1 Future Directions

Three directions remain open, all of scope.

Distributed computation: The current design targets single-node execution. Extending to distributed memory systems would enable exploration of larger state spaces than one machine’s memory admits.

Rule learning: Automatic discovery of rules producing desired structures (e.g., emergent dimensionality, causal invariants) remains an open challenge amenable to learned heuristics.

Verification beyond the bound: The properties of Section 7 are established for bounded programs under RC11 and, for the forwarding protocol, over an unbounded worker count against a transcribed model. Neither establishes the code’s properties for unbounded thread counts. A mechanised proof over the implementation itself would.

9.2 Availability

The engine is used directly as a library and as the evolution backend for downstream tools; with

BUILD_VISUALIZATION enabled it emits an event stream that an external renderer consumes to draw the evolution, causal and branchial graphs. The implementation is available as:

- Source code: <https://github.com/WolframInstitute/HypergraphRewritingEngine>
- Wolfram Language paclet: `WolframInstitute/HypergraphRewriteEngine`, installable with `PacletInstall["WolframInstitute/HypergraphRewriteEngine"]`

Acknowledgments

The author thanks Stephen Wolfram for foundational work on the Wolfram Physics Project. This work was supported by the Wolfram Institute.

References

- [1] Stephen Wolfram. A project to find the fundamental theory of physics. <https://www.wolframphysics.org>, 2020. Wolfram Media.
- [2] Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism, II. *Journal of Symbolic Computation*, 60:94–112, 2014.
- [3] Boris Weisfeiler and Andrei Lehman. The reduction of a graph to canonical form and the algebra which appears therein. *Nauchno-Tekhnicheskaya Informatsia*, 2(9):12–16, 1968.
- [4] Hartmut Ehrig, Karsten Ehrig, Ulrike Prange, and Gabriele Taentzer. *Fundamentals of Algebraic Graph Transformation*. Monographs in Theoretical Computer Science. Springer, 2006.
- [5] Grzegorz Rozenberg, editor. *Handbook of Graph Grammars and Computing by Graph Transformation*, volume 1. World Scientific, 1997.
- [6] Detlef Plump. The graph programming language GP. In *Algebraic Informatics*, pages 99–122. Springer, 2009.
- [7] Johannes Köbler, Uwe Schöning, and Jacobo Torán. *The Graph Isomorphism Problem: Its Structural Complexity*. Birkhäuser, 2012.
- [8] László Babai. Graph isomorphism in quasipolynomial time. *Proceedings of the 48th Annual ACM SIGACT Symposium on Theory of Computing*, pages 684–697, 2016.
- [9] Nino Shervashidze, Pascal Schweitzer, Erik Jan Van Leeuwen, Kurt Mehlhorn, and Karsten M. Borgwardt. Weisfeiler-Lehman graph kernels. *Journal of Machine Learning Research*, 12:2539–2561, 2011.

- [10] Wolfram Physics Project. SetReplace: A Wolfram language package and C++ library for hypergraph rewriting. <https://github.com/maxitg/SetReplace>, 2020.
- [11] Wolfram Research. Wolfram/multicomputation. Wolfram Language Paclet, 2023. URL <https://resources.wolframcloud.com/PacletRepository/resources/Wolfram/Multicomputation/>. Provides MultiwaySystem, the authority this work is compared against.
- [12] Zhengyi Yang, Wenjie Zhang, Xuemin Lin, Ying Zhang, and Shunyang Li. HGMatch: A match-by-hyperedge approach for subgraph matching on hypergraphs. In *IEEE International Conference on Data Engineering (ICDE)*, pages 2063–2076, 2023.
- [13] Ha-Nguyen Tran, Jung-jae Kim, and Bingsheng He. Fast subgraph matching on large graphs using graphics processors. In *Database Systems for Advanced Applications*, pages 299–315. Springer, 2015.
- [14] Xuhao Chen, Roshan Dathathri, Gurbinder Gill, and Keshav Pingali. Pangolin: An efficient and flexible graph mining system on CPU and GPU. *Proceedings of the VLDB Endowment*, 13(8):1190–1205, 2020.
- [15] Yangzihao Wang, Andrew Davidson, Yuechao Pan, Yuduo Wu, Andy Riffel, and John D. Owens. Gunrock: A high-performance graph processing library on the GPU. *ACM SIGPLAN Notices*, 51(8):1–12, 2016.
- [16] Xuhao Chen and Arvind. Efficient and scalable graph pattern mining on GPUs. In *USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2022.
- [17] Gramoz Goranci, Adam Karczmarz, Ali Momeni, and Nikos Parotsidis. Fully dynamic algorithms for transitive reduction. In *International Colloquium on Automata, Languages and Programming (ICALP)*, volume 334 of *LIPIcs*, pages 92:1–92:20, 2025.
- [18] Franz Baader and Tobias Nipkow. *Term Rewriting and All That*. Cambridge University Press, 1998.
- [19] Jaco van de Pol and Michael Weber. The fastest term rewrite engine. *Electronic Notes in Theoretical Computer Science*, 343:169–187, 2019.
- [20] Gian-Carlo Rota. The number of partitions of a set. *The American Mathematical Monthly*, 71(5):498–504, 1964.
- [21] Albert Atserias, Martin Grohe, and Dániel Marx. Size bounds and query plans for relational joins. *SIAM Journal on Computing*, 42(4):1737–1767, 2013.
- [22] Hung Q. Ngo, Ely Porat, Christopher Ré, and Atri Rudra. Worst-case optimal join algorithms. *Journal of the ACM*, 65(3):16:1–16:40, 2018.
- [23] Maurice Herlihy and Nir Shavit. *The Art of Multiprocessor Programming*. Morgan Kaufmann, 2012.
- [24] Maged M. Michael. High performance dynamic lock-free hash tables and list-based sets. In *Proceedings of the Fourteenth Annual ACM Symposium on Parallel Algorithms and Architectures*, pages 73–82, 2002.
- [25] Timothy L. Harris. A pragmatic implementation of non-blocking linked-lists. In *International Symposium on Distributed Computing (DISC)*, pages 300–314. Springer, 2001.
- [26] David Chase and Yossi Lev. Dynamic circular work-stealing deque. In *ACM Symposium on Parallelism in Algorithms and Architectures (SPAA)*, pages 21–28, 2005.
- [27] Nhat Minh Lê, Antoniu Pop, Albert Cohen, and Francesco Zappa Nardelli. Correct and efficient work-stealing for weak memory models. In *Proceedings of the 18th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP)*, pages 69–80. ACM, 2013.
- [28] Michalis Kokologiannakis and Viktor Vafeiadis. GenMC: A model checker for weak memory models. In *Computer Aided Verification (CAV)*, volume 12759 of *Lecture Notes in Computer Science*, pages 427–440. Springer, 2021.
- [29] Ori Lahav, Viktor Vafeiadis, Jeehoon Kang, Chung-Kil Hur, and Derek Dreyer. Repairing sequential consistency in C/C++11. In *Proceedings of the 38th ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, pages 618–632. ACM, 2017.
- [30] Leslie Lamport. *Specifying Systems: The TLA+ Language and Tools for Hardware and Software Engineers*. Addison-Wesley, 2002.

A Complexity Analysis Details

This appendix gives the bounds quoted in the body, for the two operations that dominate a run: deciding whether a produced state has been seen before, and finding the matches of a rule in a state. Both have a worst case far above their observed cost, and in both the gap is a property of the *input* rather than of the implementation—symmetry in the first case, left-hand side connectivity in the second—so each bound is stated together with what makes it attainable. Appendix B gives the per-operation costs of the concurrent structures these bounds are realized on (Table 20).

A.1 Canonicalization Complexity

For a hypergraph with $|V|$ vertices and $|E|$ hyperedges of maximum arity k , the two canonicalization roles have different cost profiles.

Weisfeiler–Leman signature. One refinement round aggregates, for each vertex, the multiset of neighbor colors: $\mathcal{O}(k \cdot |E|)$ work per round, with at most $|V|$ rounds before the coloring stabilizes, giving $\mathcal{O}(|V| \cdot k \cdot |E|)$ in the worst case and far less in practice (the partition typically stabilizes in a few rounds). WL is a sound but incomplete invariant: identical hashes are necessary but not sufficient for isomorphism.

Individualization–refinement (exact). IR interleaves color refinement with individualization of vertices in non-singleton cells, exploring a search tree pruned by discovered automorphisms. It is polynomial on instances that refinement alone resolves, but—like nauty and Traces—exponential in the worst case, with the blow-up concentrated on highly symmetric (high-automorphism) states. We claim no polynomial-time bound. Under the exact mode IR is applied to every state, and profiling attributes 22.9% of the engine’s executed instructions to it and the orbit computation beside it (Table 18). Bucketing by a cheaper hash and invoking IR only on collisions does not reduce this cost, by Proposition 3.3: a repeated bucket is consistent with a genuine duplicate, which IR must still confirm, so such a filter elides at most a fraction C/R of the invocations it precedes.

A.2 Pattern Matching Complexity

For pattern with $|L|$ edges and state with $|S|$ edges:

- Signature index lookup: $\mathcal{O}(1)$ per pattern edge
- Candidate enumeration: $\mathcal{O}(|S|/B_d)$ expected where B_d is the Bell number
- Match verification: $\mathcal{O}(|L|)$ per candidate
- Total: $\mathcal{O}(|L| \cdot |S|/B_d \cdot |L|) = \mathcal{O}(|L|^2 \cdot |S|/B_d)$

B Data Structure Specifications

Table 20: **Per-operation costs of the concurrent data structures.** Amortized where the structure grows; n is the number of elements held.

Structure	Insert	Lookup	Delete	Memory
ConcurrentMap	$\mathcal{O}(1)$ amort.	$\mathcal{O}(1)$	$\mathcal{O}(1)$	$\mathcal{O}(n)$
SegmentedArray	$\mathcal{O}(1)$	$\mathcal{O}(1)$	N/A	$\mathcal{O}(n)$
LockFreeList	$\mathcal{O}(1)$	$\mathcal{O}(n)$	N/A	$\mathcal{O}(n)$
SparseBitset	$\mathcal{O}(\log n)$	$\mathcal{O}(\log n)$	$\mathcal{O}(\log n)$	$\mathcal{O}(n/64)$